

The Enforcement Bottleneck: Judicial Capacity and the Limits of Bankruptcy Reform

Manaswini Rao

April 22, 2026

Abstract

The efficacy of bankruptcy reforms in strengthening lender rights and credit markets often depends on local enforcement capacity through courts. I analyze India’s national bankruptcy reform in 2016 using a triple difference research design, exploiting variation in time, district-level judge-to-population ratios, and firm-level default risk or capital efficiency (marginal revenue product of capital or MRPK). The reform reduced borrowing by high-default-risk firms but did not increase borrowing by high MRPK firms in districts with better judicial capacity. I find contractionary effects among high-default-risk firms consistent with credit discipline and limited positive effects among high MRPK firms, suggesting that such reforms fall short of achieving allocative efficiency.

¹Contact: manaswini.rao@gmail.com or manasrao@udel.edu, Department of Economics, University of Delaware. I am indebted to my advisors Aprajit Mahajan, Elisabeth Sadoulet, and Frederico Finan for their constant guidance and feedback through this project. I thank Olga Gorbachev, Adrienne Lucas, Alain de Janvry, Amrit Amirapu, Giorgia Barboni, Alfred Duncan, Marco Gonzalez-Navarro, Sean Higgins, Ethan Ligon, Jeremy Magruder, Matthew Pecenco, Carly Trachtman, Vaishnavi Surendra, and other seminar participants at UC Berkeley, WEFIDEV, and the University of Kent. I acknowledge the generous funding support from International Growth Centre (IGC), State Effectiveness Initiative, UC Berkeley Library, University of Delaware Lerner Grant for data acquisition. This paper was originally circulated under the title “Institutional Factors of Credit Allocation: Examining the Role of Judicial Capacity and Bankruptcy Reforms” and “Creditor Rights and Credit Allocation: Examining Bankruptcy Reforms from the Lens of Local Judicial Capacity”. All errors are my own.

1 Introduction

A large and influential literature in development economics has emphasized the role of legal institutions - particularly those protecting creditor rights - as central to strengthening financial markets and ultimately promote long-run economic growth (La Porta et al. 1998; Glaeser and Shleifer 2003; Acemoglu and Johnson 2005; Djankov et al. 2007). A well-functioning bankruptcy process enables better credit allocation and improves productive capacities (La Porta et al. 1998; Dahiya et al. 2003; Besley and Persson 2009; Hsieh and Klenow 2009; Buera et al. 2011; Gormley et al. 2018). On the policy side, many countries have spent time and effort in implementing bankruptcy reforms through changes in their legal framework. Yet, the consequences of these *de jure* reforms have been mixed (Vig 2013; von Lilienfeld-Toal et al. 2012; Ponticelli and Alencar 2016; Breza and Liberman 2017; Closset et al. 2023). Recent literature suggests that local state capacity - the *de facto* implementation and enforcement of such reforms could be critical (Giné and Love 2010; Cantoni and Yuchtman 2014; Chemin 2020; Boehm and Oberfield 2020; Alok et al. 2022). However, isolating the complex interactions between national-level reforms, local enforcement capacity, and subsequent allocation of credit to draw causal inference remains a challenge.

In this paper, I study the impact of this interaction by borrowers' ex-ante default risk and capital efficiency in the context of one of the largest legislative and policy reforms in bankruptcy protection in India (Insolvency and Bankruptcy Reform or IBC, 2016). This setting provides an ideal empirical design to test the limits of legal reforms on subsequent credit allocation for three reasons. First, the reform was a national shock that shifted power from borrowers to creditors. Second, there is substantial variation in local judicial capacity - measured by granular, district-level judge-to-population ratios - that affects enforcement processes involved in the recovery of capital. Third, firms in India face credit constraints to be on the productivity frontier (Hsieh and Klenow 2009; Banerjee and Duflo 2014; Breza and Kinnan 2021). This allows me to use a triple difference (DDD) research design that exploits variation in the timing of the reform with predetermined spatial variation in district court capacities and predetermined variation among local firms based on their default risk or capital efficiency (MRPK).¹

¹MRPK is a standard measure of firm's capital efficiency in the literature, for example, Bau and Matray (2023). Ideally, one would use marginal product of capital (MPK) but because data is mainly available for sales revenue instead of production, MRPK is used as a close proxy. These firm-level characteristics are generated based on data on credit rating and estimated MRPK before 2010. This addresses any endogeneity or anticipation concerns given the reform timeline. District judicial capacity is constructed using judge tenure data from Ash et al. 2025 for the period 2012-2016.

I find that local judicial capacity is one of the key binding constraints in resolving *enforcement frictions* of the reform but it stops short of improving allocative efficiency in the presence of continuing *information frictions*. In districts with high judicial capacity, the reform leads to credit disciplining effect: high-default risk firms reduce their borrowing and increase short-run liquidity. However, I find no evidence that the reform increased credit allocation to high capital efficiency (high MRPK) firms or their liquidity levels, even in high judicial capacity districts. These results suggest that while judicial capacity can reduce credit allocation based on an observable and incentivized dimension of credit risk (Chemin 2010; Ponticelli and Alencar 2016; Li and Ponticelli 2022; Müller 2022), it is insufficient to increase capital allocation towards more productive firms.

In India, local district courts played a direct role in bankruptcy process under the Provincial Insolvency Act, 1920, which the IBC 2016 reform superseded. The reform strengthened lender rights through a greater focus on asset restructuring and less on liquidation (in contrast to Chapters 11 and 7 of the US bankruptcy code, respectively).² The reform included a new legislative act and a series of policies by India’s central bank - the Reserve Bank of India (RBI) - such as the 2015 Asset Quality Review and 2018 circular that aimed to improve the efficacy of the new law by declaring bad debts promptly. However, in practice, district courts continue to play a complementary role, including execution of judicial orders, injunctions against asset liquidation, and challenges to orders passed by specialized bankruptcy courts (NCLT or DRTs). In fact, many liquidation orders under IBC have been stuck in implementation due to multiple ongoing litigation challenging these orders.³ As a result, local judicial capacity in district courts could be critical with respect to the *last mile enforcement* of national-level reforms, which in turn could affect the functioning of financial markets. Ignoring local state capacity could introduce misallocation of capital if lenders take institutional quality into account in their lending decisions through differences in de facto lender rights.

Empirical evidence on this interaction between local state capacity and national-level reforms is relatively mixed - on the one hand, Visaria 2009; Ponticelli and Alencar 2016; Müller 2022

²This was emphasized by Dr. M.S.Sahoo, the first chairman of the Insolvency and Bankruptcy Board of India in <https://timesofindia.indiatimes.com/business/india-business/insolvency-laws-aim-is-biz-rejig-not-recovery/articleshow/84125959.cms>

³This is also consistent with the law’s focus on restructuring rather than liquidation. A classic case study is the example of the first liquidation order under IBC, 2016 by India’s bankruptcy court (*VNR Infrastructures Limited v. State Bank of India & Ors*, CP (IB) No. 12/10/HDB/2017). This liquidation order continues to be mired in litigation due to challenges against liquidating one of the properties of the said bankrupt firm (*Leo Edibles and Fats Limited v. Tax Recovery Officer (Central), Income Tax Department, Hyderabad and Others.*, 2018 (3) TMI 1528 / W.P. No. 8560 of 2018).

suggest that better judicial capacity improves the functioning of bankruptcy process and relaxes capital constraints faced by firms to grow. On the other hand, [Giné and Love 2010](#); [Vig 2013](#); [Closset et al. 2023](#) document that strengthening creditor rights through better bankruptcy implementation could have negative implications for firms by increasing liquidity hoarding to overcome costly expectations of bankruptcy or increased concentration in debt structure, generating negative welfare consequences.

This paper addresses this dichotomy by examining firm-level borrowing and production decisions based on their ex-ante default risk or ex-ante capital efficiency to test whether local court capacity affects credit allocation after the bankruptcy reform. These outcomes are motivated by the fact that district courts enable the enforcement of debt recovery-related judicial orders rather than facilitate bankruptcy themselves. This implies that local institutional quality could plausibly affect the incentives of lenders in subsequent credit allocation, which can be tested empirically.

I use firm-level annual balanced panel data from a representative sample of formal sector firms in the CMIE Prowessdx database and construct predetermined measures of default-risk and capital efficiency to test credit disciplining and allocative efficiency. While allocation based on MRPK assumes that capital efficiency is observable and incentive-compatible with lenders, I use default-risk to examine allocations when lenders' incentives are more aligned with minimizing defaults rather than maximizing capital efficiency. I measure district-level judicial capacity as the predetermined judge to population ratio. This measure reflects the stark undersupply of judges to resolve legal cases, where the national average is 20 judges per million compared to over 100 in advanced economies like the United States. This is also a commonly used metric to measure judicial capacity, including in cross-country comparisons such as [Wittrup 2024](#). The low national-level average masks substantial variation between districts, where some districts have higher ratio than others. Since district-level judicial capacity is likely correlated with both levels and changes in district-level wealth, urbanization rates, and other potential confounders, I account for these through district and calendar year interactions in my empirical specification.

The assumption for causal identification using triple difference research design is that the outcomes of firms by default risk (or MRPK) trend similarly within subgroups of districts with higher or lower judicial capacities in the absence of the bankruptcy reform. This design relaxes the more stringent parallel trends assumptions but requires that high and low-default risk firms (or high and low MRPK firms) trend similarly within high or low judicial capacity districts even if they do not trend similarly on average. Since the bankruptcy reform was

a one-time national-level reform, applicable to all districts at the same time, it allays any concerns arising from staggered rollout designs (Ortiz-Villavicencio and Sant’Anna 2025). At the same time, this was a substantive reform to the country’s bankruptcy regime, and so it is likely that the timing of the reform could have been anticipated. For these to affect causal identification, the anticipation must vary within firms by their type across districts with differing judicial capacities. I implement the research design as a dynamic event study where I find no evidence of such anticipation behavior. Furthermore, I also include district-year and industry-year fixed effects to account for any location or industry-specific dynamics that could be affected by the reform.

This paper presents three main findings. First, I find that borrowing decreases but net short-term liquidity increases among high-default-risk firms in better judicial capacity districts following the reform. On the other hand, I do not find any clear effects on borrowing or liquidity among high-MRPK firms in similar districts. I note that a firm’s classification of default risk is not strongly (negatively) correlated with its estimated MRPK even after accounting for industry and location of the firms, making these relatively independent metrics. As a result, I rationalize these results on borrowing and liquidity by acknowledging that firms classified as high default risk are different from low-MRPK firms and that I see borrowing effects only by default risk and not capital efficiency. I interpret increased liquidity as suggesting that high-default-risk firms hold more cash at hand in districts with better enforcement after the reform. This result is consistent with the findings of Vig 2013; Closset et al. 2023, which documented that strengthening creditor rights increased liquidity hoarding among firms in a similar context with multiple market failures.

Second, I note significant contractionary effects on the economic outcomes of high-default risk firms, including their legal and interest expenditures, investment, and sales revenue. I find no effects on the expenditures or sales revenue among high-MRPK firms. These are consistent with the results on financial outcomes, where high risk firms that reduce borrowing also experience contraction, suggesting *credit disciplining* effects. While most outcomes of high MRPK firms show limited effects, I observe an increase in investments, which could indicate such firms’ capacity for long-run investment that is unobservable to lenders, suggesting continuing *information frictions*.⁴

Third, I examine judicial filings, resolutions, and pending backlog of firm-level legal cases

⁴The firm-level investment variable in CMIE Prowess dataset includes investments in both productive and financial projects. In the absence of borrowing effects, high MRPK firms could plausibly invest their retained earnings and cash from other reduced costs in long-run projects.

in district courts. I use detailed case-level data merged with firm litigant identity from [Rao \(2025\)](#). I also examine judicial outcomes of cases involving banks. The matched-firm case-level data presents a novel fact that district courts enable *execution* of existing judgment orders and complement specialized courts for debt recovery and bankruptcy even if they themselves don't directly adjudicate these matters. Overall, I find increased resolution of firm-level cases in district courts with better judicial capacity, particularly in cases involving *high-default-risk firms as defendants*. I find suggestive evidence of increased resolution of bank-level cases when banks appear as plaintiff although this result is not significant at conventional levels.

These contrasting results - credit disciplining effects without improved credit allocation towards high productivity firms - suggest that there are multiple binding constraints at play. While better judicial capacity relaxes some of the ex-post enforcement constraints of the bankruptcy reform, it does not solve the ex-ante information constraints in identifying high productivity firms. I find that the observed credit ratings are only weakly correlated with capital efficiency (MRPK), suggesting that lenders continue to face information frictions in identifying high productivity firms. These findings are consistent with a simple model of credit allocation in the presence of varying institutional quality and incentives faced by lenders (bank or loan officials). Typically, bank officers have lower incentives to declare a default and often provide additional loans to refinance older, delinquent debt. This phenomenon is referred to as "ever-greening" and is prevalent not only in India ([Chopra et al. 2020](#); [Kulkarni et al. 2025](#)) but also globally ([Jordà et al. 2022](#); [Li and Ponticelli 2022](#); [Fariae-Castro et al. 2024](#)). The reform along with local judicial capacity could reduce the pressure on ever-greening but it need not present mechanisms to reduce information constraints.

This paper contributes to two main strands of the academic literature. First, it provides evidence on the importance of local court capacity in complementing bankruptcy reforms. Literature has documented mixed evidence of the impact of legal reforms protecting lender and investors' interests in creating a diverse and competitive credit markets ([La Porta et al. 1998](#); [Rajan and Zingales 1998](#); [Djankov et al. 2007](#); [Gormley et al. 2018](#)), credit availability for firms ([Giné and Love 2010](#); [Visaria 2009](#); [Vig 2013](#); [Closset et al. 2023](#)) and subsequent implications for real firm-level decisions ([Ponticelli and Alencar 2016](#); [Amirapu 2021](#); [Alok et al. 2022](#); [Müller 2022](#); [Li and Ponticelli 2022](#); [Rao 2025](#)). This paper provides evidence that local courts matter in complementing bankruptcy reforms through credit-disciplining effect by enforcing existing judicial orders that help with debt recovery.

Second, I contribute to the literature on credit misallocation by bringing attention to the

tension between allocative efficiency and credit-disciplining. Regulatory state lending and bailout policies (Banerjee and Duflo 2014; Giné and Kanz 2018; Bau and Matray 2023), and incentives faced by banking officials (Cole et al. 2015; Fisman et al. 2017; Chopra et al. 2020; Faria-e-Castro et al. 2024; Naaraayanan and Wolfenzon 2024; Kulkarni et al. 2025), could ignore allocative efficiency (Müller and Verner 2023; Jack et al. 2023) and focus more on political or individual incentives (Khwaja and Mian 2008; Lilienfeld-Toal and Mookherjee 2016). This paper demonstrates that while a strong bankruptcy regime requires complementary investment in judicial capacity to reduce lending towards high-default-risk firms, it isn't enough for increasing allocation towards more productive firms.

Rest of the paper is structured as follows. Section 2 describes the banking sector, lender rights and enforcement mechanism in India under pre-reform and post-reform regimes. Section 3 presents a simple conceptual framework of credit allocation to draw testable hypotheses on credit allocation as a function of bankruptcy regime and local enforcement capacity. I estimate these hypotheses using the empirical strategy laid out in Section 4. Section 5 describes the datasets and variables used in the analysis. I report our results, robustness, and mechanism in Section 6, Section 7, Section 8 respectively, and conclude in Section 9.

2 Context

The banking sector in India is characterized by a dominant share of public sector banks and a small share of private banks, together known as Scheduled Commercial Banks or SCBs, which are regulated by the central bank, Reserve Bank of India (RBI). The state has always played a key role in the evolution of this sector through varying systems of controls on the sector's operation including nationalization or the dominant public ownership and controls over lending operations through priority sector lending norms, liquidity and cash reserves requirement (Demetriades and Luintel 1996). Borrowers, particularly agricultural and small firms, are also frequently bailed-out by the government prior to elections, eroding timely repayment behavior (Giné and Kanz 2018).

Non-performing assets (NPAs) in India have historically been over 2% and steadily increasing since 2012, peaking to slightly above 11% (Figure 1). The peak roughly corresponds to the timing of bankruptcy reform and other regulatory changes intended to address this problem of growing NPA.

The bankruptcy process in India until 2016 favored promoters/founders and shareholder

rights over lender rights during restructuring or liquidation proceedings. The 2016 Insolvency and Bankruptcy Code (IBC 2016) is a new consolidated law on bankruptcy process that provides for a market-based mechanism for time-bound resolution, emphasizing restructuring over liquidation, and at the same time prioritizing the rights of financial lenders such as banks in the recovery of unpaid debt from the borrower.

Figure 2 provides a timeline of the reform. Up until 2016, codes governing bankruptcy proceedings were fragmented across many statutes. Company Law, which lays down the rules governing incorporated entities, was amended in 2013 in an attempt to streamline the process. An all encompassing bankruptcy statute under IBC was tabled in the Parliament in the winter session of 2015, passed and became a law by May 2016. The first set of bankruptcy resolutions were passed in 2017. I use the year of the law (2016) to divide the time period into pre-reform period when the rights of financial lenders were weak and post-reform when the lenders were made key stakeholders in resolving debt defaults, particularly large scale corporate debt defaults, to decide on restructuring or liquidation as the appropriate next step.

There were many concurrent policies by the RBI, such as the 2015 Asset Quality Review and the 2018 guideline in promptly identifying defaults and initiating reporting towards bankruptcy processes. All of these policies aimed to make the IBC 2016 effective. Therefore, the interpretation of a change in bankruptcy regime that I examine in this paper includes not just the on-paper reform but also all complementary policies aimed at improving the resolution of stressed capital.

2.1 Role of the Judiciary in Bankruptcy

The dispute resolution institutions relevant to the bankruptcy regime in India include general courts of law, specialized debt recovery tribunals (DRTs) and specific bankruptcy courts - National Company Law Tribunal (NCLT) and National Company Law Appellate Tribunal (NCLAT) - that were set up under IBC 2016. These together are responsible for implementing the law and for providing clarity over both the process and outcomes for borrowers and lenders. Frontline district courts are the relevant institutions in the recovery of smaller debts (under INR 2.5 million) and for the execution of judicial orders. This role of local courts is particularly important as many aspects of the new bankruptcy regime are yet to be implemented in practice and to resolve concerns arising from the complex interaction between the new bankruptcy code and older laws, including procedural code (for example, the 1908 Code of Civil Procedure).

Specific debt recovery related cases are filed in local district courts with jurisdiction over de-

faulting firm location or in specialized Debt Recovery Tribunals (DRT) for larger debts. The court in which a suit should be filed is defined in the Code of Civil Procedure, 1908. Neither the lender nor the defaulting borrower has any choice over where they can file their lawsuit. The bankruptcy process itself is adjudicated by the NCLT (a centralized tribunal), where the suit can either be filed by the lender or initiated by *corporate* borrowers either through asset restructuring or liquidation as the last resort. However, bankruptcy proceedings related to individual borrowers under the Provincial Insolvency Act (1920), or liabilities from criminal offense, fraud and family disputes, personal insolvency, anti-trust and intellectual property, and property related disputes such as tenancy and eviction continue to be litigated within the district courts. District courts are also the relevant institutions that handle ancillary disputes arising from the enforcement of judgement orders from specialized laws or courts such as SARFAESI Act, 2002 (Vig 2013). As a result, these courts play a complementary role in the enforcement of lender rights under the new bankruptcy regime and their capacity could matter for the subsequent impact of the reform.⁵ Furthermore, district courts are ubiquitous - one for each administrative district - compared to only 29 DRTs across India or 16 locations of the NCLT. District courts make access to judicial institutions easier when defaulters are based in non-metropolitan districts far away from the location of DRTs or NCLT benches (see Figure A.1 for the spatial distribution of district courts in this study sample and DRT/NCLT benches).

How well the district courts function depend on the staffing levels in such courts (Rao 2025). While there is an active debate and discussion on how best to arrive at the “optimal” number of judges per court (*Imtiyaz Ahmed v. State of U.P. , 2012*), the method of allocating number of judges per court during the study period followed a rule of thumb approach of 50 judges per million people (*All India Judges’ Association case, 2002*). In practice, many courts fall short of this number and there is wide variation in the realized number of judges per capita across districts in India (see Table 1 on summary statistics). A lot of this variation is due to decentralized approach to judicial administration including budgetary and implementation support that is provided by respective state governments. While states with higher tax-to-GDP ratios could have higher number of judges per million, competing policy priorities for funding other projects/programs, delays in recruitment, and other infrastructure bottlenecks (lack of adequate courtrooms or physical space) could drive the observed variation. I account

⁵For example, I note multiple cases pertaining to a specific bankruptcy law (from prior to the 2016 reform) in district courts using the Ash et al. (2025) dataset. These cases invoke Section 17A of the SARFAESI Act, 2002, which allows firms or individuals to challenge liquidation of their assets by financial institutions in their local court. Additionally, over a million legal cases in district courts pertain to execution of past judgements/orders.

for these state and district-level dynamic considerations in allocating judicial resources in my empirical design that I discuss in detail in [Section 4](#).

2.2 Bank Officers’ Incentives

Credit misallocation could also arise from poor monitoring and incentives of bank officials. Since employees of public sector banks are tenured officials, who are subject to frequent transfers to different branch locations, their incentives to sanction loans may not align with the objective for efficient credit allocation but rather to maximize their own personal incentives such as enforceability of repayment ([Fisman et al. 2017](#)). Furthermore, initiating bankruptcy process requires lenders to make provisions for any expected losses, which may be at odds with the incentives of the bank official that depends on the extent of surplus on the bank’s balance sheet ([Kulkarni et al. 2025](#)). Acknowledging that such incentives could limit the impact of the reforms, India’s central bank, the RBI, circulated guidelines to all banks to tighten supervision and reporting of defaulted loans in a timely fashion, within a few months following the first set of bankruptcy resolution ([Kulkarni et al. 2025](#)). However, these guidelines did not change lenders’ incentives to lend more to entities with higher marginal product of capital and thus, the impact of the reform on efficient credit allocation is not apriori clear.

3 A Conceptual Framework on Credit Allocation

I sketch a simple two-period model on credit allocation in an environment where bank officials’ incentives are based on their recognition of bad loans (NPA). The status quo creates an incentive against detection and reporting of loan defaults by the bank officials, who continue to provide fresh loans to finance the past, defaulted loan (“ever-greening”). The bankruptcy reform attempts to change this incentive. This framework extends the model in [Banerjee and Duflo \(2014\)](#) by making ever-greening by a bank official a function of both the bankruptcy regime (de-jure *incentive*) and local judicial capacity (de-facto *enforcement*).

Consider a representative bank official, who has L units of credit to allocate to borrowers in each period. Borrowers are of two types in the population - a type H with a share ϕ in the population and a type L constituting the remaining $1 - \phi$. The type H borrower has a deterministic production function with probability of success, $s_H = 1$ whereas the type L face a non-zero probability of production failure $1 - s$ such that $s_L = s < 1$. The borrower type is unknown to both bank official and borrower in the initial period, which gets revealed with the outcome of production. So in the first round of allocation, the official equally distributes

total credit L equally among the borrowers such that each borrower gets l units of credit.

At the end of period 1, the official receives a signal that the borrower is of type H conditional on observed production success. This signal probability is $S_H = \frac{\phi}{\phi + (1-\phi)s}$. Among type L borrowers, $(1 - \phi)(1 - s)$ fail. At the beginning of period 2, the official can either declare loan provided to type L as default and face a penalty $P_1(l, \Gamma)$ or provide fresh loan to type L borrower to continue production, which would lead to a bigger default at the end of period 2 with probability $1 - s$. Penalty in period 1 is a function of loan size, l , and local court quality Γ , such that $\frac{\partial P_1}{\partial \Gamma} > 0$, i.e. well-functioning courts generate a larger penalty, and $\frac{\partial P_1}{\partial l} > 0$. The official will bail out type L borrower by lending l_L in period 2 if $sf(l_L - l) \geq l_L$, where l due from period 1 is paid out of l_L and the rest put into production. Rationality implies that $l_L^* = sf(l_L^* - l)$, that is, the official lends the minimum amount to type L borrower to hedge the risk of default at the end of period 2. Therefore, the official compares the option of bailing out at the beginning of period 2 to declaring a default, which costs him $P_1(l, \Gamma)$. So, bailing will be the dominant strategy if the penalty from default at the end of period 2 is less than the penalty from default at the end of period 1. Penalty in period 2 has an additional institutional parameter, Θ - a measure of creditor (bank's) rights in the form of redressal through bankruptcy process. I assume that $\frac{\partial P_2}{\partial \Gamma} > 0$, $\frac{\partial P_2}{\partial \Theta} > 0$ and $\frac{\partial^2 P_2}{\partial \Gamma \partial \Theta} > 0$. That is, penalty in period 2 is higher due to the complementary effects of default detection through courts as well as protection of creditor rights under bankruptcy reform. Thus, bailing will be the optimal response if the following holds:

$$\begin{aligned} (1 - s)P_2(l_L^*, \Theta, \Gamma) &\leq P_1(l, \Gamma) \\ P_2(l_L^*, \Theta, \Gamma) &\leq \frac{P_1(l, \Gamma)}{1 - s} \end{aligned}$$

Assuming that the penalty can be expressed as a share of the loan amount, $P_1(l, \Gamma) = lP_1(\Gamma)$ and $P_2(l_L^*, \Theta, \Gamma) = l_L^*P_2(\Theta, \Gamma)$. Therefore, the bailout loan can be expressed as:

$$l_L^* \leq \frac{lP_1(\Gamma)}{(1 - s)P_2(\Theta, \Gamma)}$$

The right hand side of the above expression for period 2 bailout loan varies by both the quality of local courts, Γ , as well as the overall environment of creditor rights Θ . The bankruptcy reform increases Θ whereas higher judicial capacity (higher judge to population ratio) increases Γ . From this set-up, I get the following results with corresponding hypotheses

to test with data.

Result 1: The marginal reduction in bailout loan, l_L^* , is amplified by increasing institutional parameters Θ and Γ , i.e. $\frac{\partial l_L^*}{\partial \Theta \partial \Gamma} < 0$. That is, bailout lending to type L borrowers decreases in credit markets (districts) with higher judicial capacity after the bankruptcy reform.

Proof ⁶

$$\frac{\partial l_L^*}{\partial \Theta \partial \Gamma} = \frac{l}{1-s} \left(\frac{-1}{P_2^2} \frac{\partial P_1}{\partial \Gamma} \frac{\partial P_2}{\partial \Theta} - \frac{P_1}{P_2^2} \frac{\partial^2 P_2}{\partial \Theta \partial \Gamma} + \underbrace{\frac{2P_1}{P_2^3} \frac{\partial P_2}{\partial \Gamma} \frac{\partial P_2}{\partial \Theta}}_{\rightarrow 0} \right) < 0$$

Result 2: In the absence of strong creditor rights through the bankruptcy reform, the effect of higher judicial capacity on bailout loan is ambiguous or likely positive.

Proof

$$\frac{\partial l_L^*}{\partial \Gamma} = \frac{l}{1-s} \left(\frac{1}{P_2} \frac{\partial P_1}{\partial \Gamma} - \frac{P_1}{P_2^2} \frac{\partial P_2}{\partial \Gamma} \right) > \text{ or } < 0$$

Implication Given this framework, the following section presents the empirical strategy to test the hypotheses on credit allocation generated by the above results. Specifically, from result 1, I examine whether lending to “riskier” borrowers (from the perspective of default probability) diminishes after the bankruptcy reform in districts with better judicial capacity. Result 2 suggests plausible “parallel trends” prior to the reform since lending decisions to specific type of borrower are more likely to be driven by interaction with bank official’s incentives rather than institutional quality alone.

4 Empirical Strategy

I follow a triple difference (DDD) research design to test the hypothesis from Result 1 from the framework above that local judicial capacity plays a complementary role to bankruptcy reforms in reducing lending to risky borrowers. As a corollary, I also examine allocation by capital productivity to test for allocative efficiency. I estimate the following specification using predetermined variation in firms’ default risk and variation in local judicial capacity

⁶The cubic for a large positive penalty P_2 is orders of magnitude large relative to P_1 , which makes the ratio a very small number such that the third term in the expression tends to 0 in limit.

measured as judge to population ratio. To analyze the impact on the efficiency of credit allocation, I estimate a similar specification but using predetermined variation in firms' estimated MRPK.

$$Y_{fidt} = \beta_1 \text{Hi Risk}/\text{MRPK}_f \times \text{Hi-Jdg-Pop}_d \times \text{Post}_t + \beta_2 \text{Hi Risk}/\text{MRPK}_f \times \text{Post}_t + \delta_f + \delta_{it} + \delta_{dt} + \epsilon_{fidt} \quad (1)$$

where Y_{fidt} represents relevant outcomes (borrowing or productivity) of firm f , in industry i , registered in district d in year t . Firms are classified as high-default-risk or high-MRPK as defined in [Section 5](#), where Hi Risk_f is a dummy variable taking value 1 if their credit rating falls in any of the 3 risk categories at least once before 2010. Similarly, Hi MRPK_f is a binary variable based on categorization of firms as above median in the distribution of pre-2010 estimated MRPK across the sample firms (coded as 1 and below median coded as 0). Districts are classified as high judicial capacity, which is denoted by dummy variable High Jdg Pop_d that takes value 1 if the judge-to-population ratio in the 2012-2016 period is above the sample average of 51.7 judges per million, and 0 otherwise. This threshold also maps to the recommended staffing levels by the Supreme Court of India. Post_t is a dummy indicating whether year $t > 2016$ (i.e. period after the reform), defined as $\text{Post}_t = 1$, for $t > 2016$ and 0 otherwise. Since the data follows a panel structure, I include firm fixed effects denoted by δ_f to account for all firm-level, time invariant unobserved potential confounders. In addition, I flexibly account for unobserved dynamic confounders at industry and local district levels using industry-year (measured as 2 digit NIC product code associated with each firm) and district-year interactions. These are denoted by terms δ_{it} and δ_{dt} , respectively. ϵ_{fidt} is the idiosyncratic error term. Since the “treatment” varies at the district level, I cluster standard errors by district.

The main null hypothesis of interest is whether the coefficient $\beta_1 = 0$ in [Equation 1](#) above. A negative and significant coefficient would imply that the outcome declines in the post reform period in districts with better judicial capacity for a firm belonging to the specific category. Similarly, a positive coefficient would imply that the outcome increases in the post reform period in district with better judicial capacity for firms of a specific type. Result 1 from [Section 3](#) implies that $\beta_1 < 0$ for high-default-risk firms. Moreover, if we expect allocative efficiency to improve (which is not explicitly modeled), then we expect $\beta_1 > 0$ for high MRPK firms.

To interpret β_1 as a causal parameter, the assumption is that borrowing (and any other out-

comes, Y_{fidt}) by hi-default-risk (high MRPK) trends in parallel to those by low risk firms (low MRPK) within high judicial capacity or low judicial capacity districts, even if, on average, they don't trend in parallel to each other. That is, the design relaxes the assumption that high-risk firms trend similarly to low-risk firms everywhere. For example, it is plausible that high-risk firms could experience more unobserved shocks over time due to their behavior, which could put them on a different trend than low-risk firms. While this would be problematic in a difference in differences (DiD) design, the DDD design allows such differential trends as long as the same trends are observed in high and low judicial capacity districts. Furthermore, I include industry-year and district-year fixed effects in order to control for any unobserved time-varying confounders by industry or geography, such as variation in resource allocation to district courts over time. Finally, because the bankruptcy reform was a one time, pan-India reform with no staggering of treatment, I do not impose additional restrictions of homogenous treatment effects across space and over time.

I implement Equation 1 using two-way fixed effect specification (firm and year fixed effects) with industry-year and district-year interactions, defining post-period to include years following the reform in 2016. I also implement the design using an event study specification, presenting β_1 for each year-bin prior to and post reform. The event study implementation allows examining the dynamic effects of the reform across varying judicial capacity by firm type and also test for the parallel trends assumption within district groups by judicial capacity.

4.1 Testing Identifying Assumptions for Causal Inference

Since the reform was a one-time, pan-India reform, all districts and firms experienced the regime change at the same time. This allays any concern arising from staggered roll-out research designs that require strong assumptions for causal identification. However, it is plausible that local actors such as banks could lobby to improve their respective courts' judicial capacity by asking for more judges in areas with more efficient or high risk firms in order to initiate bankruptcy processes. This would make judicial capacity endogenous to the bankruptcy regime. I address this concern by using pre-reform judge to population ratio as a pre-determined measure of local institutional capacity to test my main hypotheses.

I test for the assumptions for causal inference in two ways. First, I examine whether there are any differential trends in outcomes between high risk/MRPK firms and low risk/MRPK firms across districts with varying judicial capacities before the reform (i.e., look for any "pre-trends" in the pre-reform period in the DDD set up). Second, I estimate and examine

the prior-period trends in a simple DiD framework by examining outcomes across all firms between high and low judicial capacity districts before the reform. This is to test whether the composition of firms and their dynamics prior to reform could confound the post reform effects in the DDD specification.

Another concern is contemporaneous expansion of bankruptcy courts (NCLT and its regional benches) following the 2016 reform, which could confound whether the results are driven by local district court capacity. This concern is effectively addressed due to the specific context of this study as the newly introduced bankruptcy courts had a more aggregate jurisdiction than that of a district court (it was even larger than a state). At the time of the reform in 2016, the federal government introduced 10 regional benches and one principal bench (New Delhi), which was slowly expanded to 16 benches in total as of 2026. Since the geographic scale of these benches is much larger than a district, the empirical specification [Equation 1](#) exactly accounts for such concerns through the district-year fixed effects. Consequently, the observed estimates are to be inferred as additional effects of local court capacity over and above any simultaneous effects of new bankruptcy courts.

Furthermore, inclusion of district-year fixed effects when the outcome is measured at a firm-year level also addresses concerns about potential, time-varying unobserved confounders like district wealth levels or urbanization trends in driving better judicial capacity over time. The main assumption for causal identification, therefore, is that there are no *differential* trends in how economic agents like banks respond to firms by their default risk or capital productivity in districts with better or worse judicial capacity independent of the bankruptcy reform or its enforcement. I show the validity of this assumption through multiple falsification tests that I discuss later in [Section 6](#).

4.2 Effect on Litigation Outcomes

Since one of the key assumptions in our model in [Section 3](#) is that institutional quality Γ matters in credit allocation, I use legal case-level data, merged with firms' identity to examine how local courts resolve cases pertaining to firms of specific type (high-default-risk or high-MRPK) or those pertaining lenders (banks) before and after the reform. Finally, as the case records allow me to identify the litigant type of firm involved in the case - as plaintiff or defendant, I estimate the following specification to examine firm-specific litigation outcome L_{fidt} - number of cases filed, resolved, or pending - separately by litigant-type. All subscripts and RHS variables are as defined before.

$$\begin{aligned}
L_{fidt} &= \gamma_1 \text{Hi Risk/MRPK}_f \times \text{Hi-Jdg-Pop}_d \times \text{Post}_t + \gamma_2 \text{Hi Risk/MRPK}_f \times \text{Post}_t \\
&+ \delta_f + \delta_{dt} + \delta_{it} + \epsilon_{fidt}
\end{aligned} \tag{2}$$

For banks, I estimate a DiD specification instead of a DDD to examine bank-specific litigation outcomes, separated by litigant-type (for cases where banks appear as a plaintiff or as defendant), to examine how debt recovery cases are handled by courts of differing capacities before and after the bankruptcy reform. This slightly different approach is required since the same classification of firms as high risk or high MRPK does not apply to banks. Moreover, banks are special types of firms that operate through their location-specific branch offices to make lending decisions. So, even for a national bank, local institutional quality would matter because their lending decisions are made by their local branch offices. Using the specification below, I examine the effect of high judge-to-population ratio of a district's judiciary Hi Jdg Pop_d on banks' b litigation outcomes L_{bdt} - number filed, number resolved, number pending - in district d in year t , before and after the bankruptcy reform after accounting for bank fixed effect δ_b , district fixed effect δ_d , and year fixed effect δ_t .

$$L_{bdt} = \gamma \text{Hi Jdg Pop}_d \times \text{Post}_t + \delta_b + \delta_d + \delta_t + \varepsilon_{bdt} \tag{3}$$

As before, I implement [Equation 2](#) and [Equation 3](#) using two-way fixed effect specification pooling all the post-reform period as well as using an event study design to present the dynamic effects of the reform.

5 Data

I use two main datasets: one to measure judicial capacity and judicial outcomes, for which I use [Ash et al. \(2025\)](#) and [Rao \(2025\)](#) datasets, respectively. The second to measure firm-level characteristics to study allocation and primary outcomes of interest - borrowing, short-run liquidity, and production outcomes, for which I use CMIE Prowess dx dataset. I merge the two datasets by the district identifier of firms' registered office. I describe both the data and variables constructed for analysis in greater detail below.

5.1 Judicial Capacity and Judicial Outcomes

Judge to population ratio is a standard measure of judicial capacity used in the literature (see [Wittrup 2024](#); [India Justice Report 2025](#)). More broadly, staff-per-capita is a commonly used metric to reflect underlying state capacity in the absence of data on service quality or efficiency metrics. For example, [Chalfin and McCrary \(2018\)](#) show that number of police officers per capita is a strong determinant of crime reduction in large cities in the United States. [Needleman et al. \(2002\)](#) discuss how shortage of nurses affects health outcomes and mortality of patients. In the context of India, [Dhaliwal and Hanna \(2017\)](#) show that increased attendance of nurses in primary health centers reduced children born with low birth weight. More directly related to the context of this paper, [Rao \(2025\)](#) shows that backlog in district courts reduces when more judges are added to a court.

I use the universe of legal case data from [Ash et al. \(2025\)](#) to calculate district-level judicial capacity measured as judge-to-population ratio. I use the judge data table to compute total number of judges observed prior to the reform using judge tenure start and end dates. The data table includes the following variables: state, district, court establishment (or simply the court), and judge position. For each judge position within a court, the data also contain the start date and the end date of the judge assigned to the position.⁷ Next, I create a dummy variable assigning a value 1 for every calendar year between the start and end years for the assigned judge, and 0 for calendar years outside this range. I do this for all of state-district-court-judge position-start year-end year tuples over 2012-2016 to calculate the maximum active judge positions in the respective district and normalize relative to the 2011 district population. Finally, I define districts with above average value (51.7 judges per million) as “high judge population ratio” districts. This also maps to districts that are above the recommended staffing levels by the Supreme Court of India.⁸

To measure litigation-specific outcomes such as filing rates, resolution rates, and pending cases at firm-level, I use data from [Rao \(2025\)](#) containing all meta-data and identifying information from the universe of legal case records from a sub-sample of 195 districts over the same time period. I match firm-level identifiers with the legal case-level data from this dataset using firm name and place of registration, followed by manual verification of the resulting match. I identify over 3500 unique non-financial firms with cases across these courts with different baseline levels of judge-to-population ratios. The metadata allows me

⁷The public version of the data does not contain any identifiers such as judge or litigant names.

⁸The average judge to population ratio based on this approach is 18.6 judges per million using all districts, which is very close to the aggregate number of 20 reported by the Supreme Court of India. The average ratio for districts in the study sample, which correspond to the matched firm-location is higher at 51.7.

to tag each firm-case record to identify whether the firm appears as a plaintiff or as a defendant. For each combination of firm-court-litigant type, I calculate number of new cases filed, number of cases resolved, and total pending backlog using filing and decision time stamps in the meta-data. I follow similar algorithm for cases involving banks and financial institutions.

5.2 Firm-level Data

I use CMIE Prowess dx dataset that contains a representative sample panel data of corporate entities (registered formal sector firms) in India with annual financial and production data since 1990s. This dataset offers the advantage of a large-scale firm-level panel data for India, which includes company name, registration details in addition to annual financial and production information. It represents multiple sectors including manufacturing, trade, and services, and thus suitable for research questions requiring firm-level information and their fine-geographic footprint. I am interested in non-financial sector firms, which form the main sampling frame for our analysis. I merge these firms with the district-level judicial data by firms' registered office location to generate the analysis sample.⁹

For primary outcomes, I focus on firms that were incorporated prior to 2010 and continue to exist through the study period. This allows me to construct a balanced panel of incumbent firms with annual outcomes spanning 2010-2024, while minimizing concerns about compositional changes to the sample of firms over time. The main analysis sample contains a balanced panel of 3500 firms located across 243 districts in India.¹⁰

In order to examine credit allocation across the sample firms, I classify firms based on their default risk and capital productivity from credit ratings and estimated MRPK using data prior to 2010, respectively. Using pre-2010 data introduces a gap of at least 5 years between measurement and the reform, which could potentially address anticipation concerns in the context of a large-scale, national-level reform.

To measure default risk, I use firms' historic credit rating information. Each year, firms are

⁹Other firm-related datasets from India are the Annual Survey of Industry (ASI) and the Micro-Small and Medium Enterprise (MSME) datasets, neither of which provide firm-level identity and their specific geographic location. However, the key trade-off in using Prowess is that it excludes smaller firms and those in the informal sector. Information from these can mainly be collected through primary enterprise surveys, often requiring large funding for data collection.

¹⁰To address concerns about composition of firms in the sample, particularly due to firm exits, I also examine survivorship of firms following the reform by underlying judicial capacity. Since I do not observe actual firm liquidations and exit in the data, I use non-reporting or disappearance in Prowess sample as a proxy.

rated based on their past performance on credit contracts by credit bureaus. There are five credit bureaus - CRISIL, ICRA, CARE, IND-RA, and BRICKWORK. These are similar to global agencies like Moody’s and Standard & Poor’s. Ratings like {PR 1+, A1+, A1, PR2, A+, A, A-, ... }, which map to specific credit safety measures are classified into 8 bins in decreasing order as “Highest Safety, High Safety, Moderate Safety, Adequate Safety, Inadequate Safety, High Risk, Substantial Risk, and Default”. I classify firms as high-default-risk if their credit rating in the pre-reform window 2005-2010 is marked “High Risk” or worse (i.e., High Risk, Substantial Risk, or Default).

I estimate firm-level MRPK by assuming Cobb-Douglas revenue function as in the literature (for example, [Bau and Matray 2023](#)) to estimate input elasticities and then construct the marginal revenue product of capital. This is a two-step process where in the first step, I estimate the input shares in the revenue function using panel regression of log sales on log capital (value of plants and machinery) and log labor (wagebill) with firm and year fixed effects. In the second step, I take the estimated coefficients on capital as the common output elasticity ($\hat{\alpha}$) and compute firm-year average revenue product, $(\frac{Y}{K})$. Finally, I define $MRPK_{it} = \hat{\alpha} \cdot (\frac{Y_{it}}{K_{it}})$. This yields firm-level measures of MRPK used for classification and heterogeneity analysis. Since this estimated measure is a continuous variable, I classify firms as high-MRPK if their estimated MRPK in the pre-reform window 2005-2010 is above the median value across all sample firms. Panel A [Figure A.10](#) shows similar distribution of estimated MRPK across all sample firms and firms in the manufacturing sector where capital is more reliably measured.

5.3 Outcome Variables

The first set of primary outcomes include borrowing and short-run liquidity (net working capital) to test our specific hypotheses on credit allocation. The second set of outcomes include economic outcomes measured as legal and interest expenditures, investments, and sales revenue to examine productivity implications of the reform and judicial capacity. I impute any missing values of the outcome measures using the firm-specific average during the study period.¹¹

Since I do not observe actual firm exits in the data, I use non-reporting or disappearance due the construction of the balance panel as a proxy to examine whether the reform differentially changes the composition of firms by underlying district judicial capacity. Similarly, I also

¹¹This approach is preferred over other methods such as industry-level imputation to preserve firm-level heterogeneity that is central to causal identification in this paper as also followed in [Link et al. 2024](#) for corporate investment variables. To ensure robustness, I examine selection into reporting in the data.

examine “balance” using selection into reporting to ensure that the results are not driven by compositional changes.

5.4 Summary Statistics

Table 1 presents a summary of the court-firm matched dataset used in the analysis. The average judge to population ratio in the sample is around 52 judges per million. This reflects the spatial distribution of formal sector firms in India rather than the population of all district courts in India. Second, the sample firms are large, formal sector firms, where 80% of the sample are publicly listed on various stock exchanges. These firms exhibit a large variation in their default risk (average rating close to 6) and capital efficiency estimated as MRPK (average marginal return of 9 for every additional rupee of capital). Most firms are between 20-60 years old, 60% engaged in manufacturing and the remaining in trade and construction. Many of the firms in the Prowess dataset frequently are involved in litigation in district courts (see Figure A.2). Table A.1 compares firms by default risk and capital efficiency using pre-2010 data across districts with high and low judge to population ratios. As expected, there is sorting of firms spatially, with better firms located in high judge-population districts on average. Interestingly, there is limited observed differences between high-default-risk firms across districts with differing judicial capacities.

6 Results

6.1 Borrowing and Liquidity

I start my analysis with borrowing and short-run liquidity outcomes to examine credit discipline - how default risk is handled - and allocative efficiency in response to change in bankruptcy regime and local enforcement capacity through district courts. Since default risk is typically observable as corporate credit ratings, I first estimate Equation 1 using default-risk classification. Panel A Figure 3 (left) presents the event study estimates and Col 1 Table 2 presents the DDD estimates. I find a reduction in borrowing by high risk firms in better judicial capacity districts, which persists in the long run (about 30% reduction). The post-reform average treatment effects on the treated (ATT) is about -16% (Col 1 Table 2).

An improvement in allocative efficiency would suggest that lenders should prioritize firms with higher marginal product of capital (or its proxy MRPK). However, I don’t find evidence in support of this and I am unable to reject the null hypothesis of no effect on borrowing by

firms' MRPK in better judicial capacity districts after the reform. Both Panel A [Figure 3](#) (right) and Col 3 [Table 2](#) suggest muted effects on borrowing by high-MRPK firms.

These results raise concern whether the overall credit market contracted. That is, since borrowing by high-risk firm decreased without a commensurate increase in borrowing by high-MRPK firms, what really happened to credit supply and demand in the local markets? While the primary triple difference strategy is geared towards isolating relative credit allocation by firm type within a balanced panel, it does not directly capture overall changes in borrowing across all firms within a district.

To answer this, I estimate the effects on overall borrowing for the average firm using a standard difference in differences (DiD) strategy in [Figure A.3](#). I find that the average firm-level borrowing increased in better judicial capacity districts post reform. This suggests that credit supply “released” by high-default risk firms was plausibly recirculated to safer and mid-productivity firms rather than to high-MRPK firms.

In contrast to the negative effects on borrowing, I find that high-default risk firms increase their short-term liquidity measured as net working capital. Panel B [Figure 3](#) and Col 2, Col 4 [Table 2](#) present the event study and DDD estimates. I find a persistent increase in short term liquidity by over 80% among such firms in better judicial capacity districts post reform. Not only do high-risk firms borrow less, they are more likely to hold cash at hand as also documented as a consequence of improved creditor rights in the literature ([Vig 2013](#); [Closset et al. 2023](#),). On the other hand, I find muted and statistically insignificant effects among high-MRPK firms in better judicial capacity districts (Col 4 [Table 2](#)).

I interpret these results within a context of limited overlap between the sets of high default-risk and low MRPK firms. I find weak negative correlation between a firm's classification as high-default-risk and its classification as high-MRPK. Without any fixed effects, a high-default-risk firm is 9% less likely to be classified as a high-MRPK firm (Col 1 [Table 3](#)). After adjusting for industry or location or both, this correlation coefficient halves and loses statistical significance. As a result, any information on default risk, such as that provided by credit rating, is less informative about a firm's capital efficiency or MRPK. This would be surprising in a world with perfect information where we would expect that high-MRPK firms should be more profitable and thus, less likely to default. However in a context with multiple factor market failures, this may not be surprising. For example, consider firms that have made huge, upfront investment and that expected future productivity includes some risk.

Such firms are susceptible to business cycle shocks, which could lead them to default given their relatively weak cash position. Another exactly opposite example would be incumbent firms with moderate or even low MRPK but with consistent cash flow such that they are less likely to default. Both these examples illustrate hidden type and hidden action mechanisms that are not currently accounted in the computation of credit ratings, which is mainly based on observable and backward-looking measures such as leverage and past payment history. The positive liquidity effect is a plausible strategy to pay back any outstanding dues to avoid initiation of bankruptcy proceedings.

Thus, while better courts could lead to some improvement in credit allocation post reform - borrowing by high-default-risk firms decline, indicating a plausible credit disciplining effect, the reform also falls short of improving overall allocative efficiency based on capital productivity.

6.2 Economic Outcomes

Figure 4 and Table 4 presents the estimates on economic outcomes from the DDD design both as event study and two-way fixed effects specification, respectively, comparing firms' by their baseline default-risk across districts with varying judicial capacities. I find significant decline in such firms' legal (17%, $p < 0.01$) and interest expenditures (33%, $p < 0.01$) in districts with better judicial capacity. I also find reduced investment by such firms, at least in the initial years following the reform, and a negative but noisy effects on their sales revenue.

Figure 5 and Table 5 presents the results using firms's baseline MRPK across districts with better or worse judicial capacities. Consistent with credit outcomes, I find insignificant effects on legal or interest expenditures or on sales revenue. On the other hand, I find a significant increase in investment by such firms after the reform in better capacity districts. The investment effect could be driven by retained earnings by these firms or through other sources of financing such as trade credit in the absence of borrowing effects.

While I don't observe firm exits or liquidation in the data, the estimated effects among high-default risk firms are consistent with *credit disciplining* effects. Such firms could also continue to survive through restructuring as opposed to complete liquidation and exit. Increased investment by high-MRPK firms is plausibly consistent with *persistent information frictions*, where high-MRPK firms that are better at making long-run investments are essentially unobservable to lenders for an impact on their borrowing outcomes.

7 Robustness of the Results

7.1 Firm Survivorship

Since the main analysis uses balanced panel of incumbent firms that survive the sample period, it does not address whether the bankruptcy reform and better local judicial capacity increased firm exits through liquidations.¹² Firm exits could affect how the sample firms respond to the underlying business environment beyond factors accounted for in district-year fixed effects and thus, could affect the interpretation of the estimates. I use absence of reporting in Prowess data as a proxy for exit since I do not observe firm liquidation in the data. [Figure A.4](#) and [Table A.2](#) present the patterns in non-reporting among all Prowess firms as well as by their default risk/MRPK. The results suggest that if anything more firms report their financial data (or fewer missing observations) in high judicial capacity districts post reform. In particular, high risk firms are more likely to report their financial data. These positive reporting effects could plausibly indicate asset restructuring instead of complete exit. I find similar lack of effects when I examine patterns in selection into reporting ([Figure A.5](#) and [Table A.3](#)).

7.2 Role of Bank-Firm Relationship

The dataset used in the main analysis is firm-level and therefore, credit allocation behavior is inferred from observed borrowing and liquidity by sample firms over time. To examine bank-side action on allocation, I make use of firm-bank relationship reported in the “bankers” table in Prowess, which reports all banks, including commercial banks, regional banks, and credit unions, which a firm has had a banking relationship with. I generate two variables to examine the allocation patterns: First, whether there exists a relationship between a firm and a bank and second, total number of banking relationships by a firm. [Figure A.6](#) and [Table A.4](#) present the results from analyzing this relationship as a proxy for bank-side allocation decision using the triple difference specification in [Equation 1](#) ([Figure A.6](#) and [Col 3-6 Table A.4](#)). Further, because I have the identity of the banking partner for two specifications, I also include bank fixed effects as indicated in the columns.

This exercise indicates that there are no “extensive” margin changes in banking relationships by either high-default-risk firms or high-MRPK firms. The results on reduced borrowing amounts suggests that the relevant margin of action is the intensive margin, where banks may lend lower amount of capital to high-default-risk firms even if they do not eliminate

¹²If a firm is restructured, they continue to exist but with different structural adjustments to their financial and business management.

their relationship with such firms entirely. Again, this is consistent with the focus of the bankruptcy reform on restructuring rather than complete liquidation.

7.3 Concurrent Shocks

In addition to the 2016 IBC code, there were many other concurrent policy shocks that could have differentially affected firms by their default risk or MRPK across high or low judicial capacity districts. Two significant shocks were demonetization launched in November 2016 and the roll out of new value-added tax regime (GST) in early 2017. Demonetization created cash shortages everywhere and it is plausible that high risk firms could have been particularly affected. To threaten identification, the shock needs to differentially affect high-risk/high-MRPK firms in high judicial capacity districts. I address this in two ways: First, I exclude 2016-2017 time period from my event window since demonetization and GST roll-out were temporary shocks, and find that the borrowing and liquidity effects by firm type persist in the long run (see [Figure A.7](#)). Second, since these shocks were felt more at an industry-level where all firms within specific industries were more affected than those in other industries, I account for these industry-specific dynamic trends through industry-year fixed effects.

7.4 Other Threats to Identification

In all main event study implementation of the triple differences research design, the pre-period trends are flat. I also test for pre-trends in the primary outcomes among all-firms using a DiD design. I consistently note lack of pre-trends on borrowing as well as economic outcomes among all firms prior to the reform ([Figure A.3](#), [Figure A.8](#), [Table A.5](#)).

I carry out additional tests using inframarginal firms as placebo in the triple difference specification, where firms are defined as inframarginal if their pre-reform credit rating is perfect. Borrowing or liquidity holding by such firms should not be affected by local institutional (district court) capacity and/or changes in lenders' incentives since these firms never default. [Figure A.9](#) presents the results from this analysis. I am unable to reject the null on borrowing or liquidity effects among these firms by district judicial capacity before and after the reform, suggesting that credit allocation likely changed for specific types of firms.

7.5 MRPK Estimation

One concern about the null effects in MRPK triple differences estimation is whether the null is due to attenuation bias from measurement error in the estimated MRPK. This could arise if the valuation of plants and machinery used as capital stock is noisily reported in data.

I address this in two ways: First, I show that the kernel density distribution of estimated MRPK for all firms across sectors as well as those in manufacturing - where capital stock is better measured, share the same support. The adjusted and within- R^2 of 0.93 and 0.54, respectively, suggests that the production function explains more than half of the within-firm variation in sales revenue (see Panel A [Figure A.10](#)). Second, I estimate [Equation 1](#) on the subsample of manufacturing sector firms, where capital measurement is relatively more reliable, to find similar patterns: the reform does not appear to increase borrowing and liquidity among high-MRPK manufacturing firms in high judicial capacity districts (Panel B).¹³ Finally, it is plausible that borrowing by high-MRPK firms could increase on average but because of spatial concentration of such firms, local judicial capacity has limited complementarity with the bankruptcy reform.

7.6 Alternate Definitions of Judicial Capacity

These results are not driven by how I define judicial capacity. I find similar results if I use median value of pre-reform (2012-2016) judge to population ratio to classify as high or low judicial capacity instead of the average (see [Table A.6](#), [Table A.7](#), [Table A.8](#)). The choice of using the average is motivated by the Supreme Court of India’s recommendation on judge-to-population ratio of at least 50 judges per million. I also use pre-reform vacancy rates among district judge positions as another way to measure judicial capacity. I find qualitatively similar effects of the bankruptcy reform in high judicial capacity districts (defined as below median vacancy rates in the pre-reform period) on firm-level outcomes including borrowing, liquidity, and production outcomes (see [Table A.9](#), [Table A.10](#), and [Table A.11](#)). Lastly, it is plausible that judicial capacity measured prior to the reform could itself change over time if capacity constraints in other districts are eventually resolved. These district-level dynamics are accounted in district-year fixed effects, which reduces concerns about measurement errors arising from changing district judicial capacities over time.

8 Mechanism: Last Mile Enforcement by District Courts

I use legal case data matched to firm identifiers from [Rao \(2025\)](#) to examine the role of local district courts. Three facts are relevant to this paper: First, the sample district courts spans and is larger than the set of districts with specialized tribunals like DRT ([Figure A.1](#) - there are only 25 districts with debt recovery or bankruptcy courts in contrast to one district court

¹³There is an immediate dip in borrowing observed for both placebo and high-MRPK manufacturing firms in [Figure A.9](#) and Panel B [Figure A.10](#) respectively. These correspond to years with cash shortages including 2016 and 2017, which revert to the mean of no differential effect in the long run.

per district). Second, many of the Prowess firms across sectors routinely use district courts, with multiple cases per firm (see Panel B [Figure A.2](#)). Third, the role of district courts is in the *enforcement* of judicial orders through execution petitions and cases that are relevant to bankruptcy enforcement such as suits for possession of collateral (see [Figure A.11](#)).¹⁴

I examine three main judicial outcomes specific to the matched firms - number of cases filed in the district courts, number of cases resolved, and number of pending cases - for each firm in the analysis sample during the study period. Next, I estimate the main identifying specification [Equation 2](#) using these firm-level judicial outcomes as the dependent variables, separately when the sample firm appears as a plaintiff and a defendant.

Panel A [Figure 6](#) and Col 1-3 [Table 6](#) present the results for high-default-risk plaintiff firms and Panel B [Figure 6](#) or Col 1-3 [Table 6](#) present the results for corresponding high-risk defendant firms. The coefficient on Hi-Jdg-Pop x Post suggests that better district courts have higher rates of resolution and lower pending backlog per defending firm in the post reform period. These court-level effects are muted for plaintiff firms in the post reform period. The DDD event study parameters (Panel B [Figure 6](#)) are noisy but suggest that more cases are filed against high default-risk firms (i.e. as defendants) and there is increased resolution of cases against such firms in courts with better judicial capacity. On the other hand, there is no conclusive evidence on judicial outcomes for cases involving high-default-risk firms as plaintiff.¹⁵

[Figure 7](#) and [Table 7](#) presents the results for plaintiff and defendant high-MRPK firms litigating in the sample courts. The coefficients on high judge-population ratio in the post reform period suggests higher rates of resolution and lower pending backlog per firm against all defending firms. The event study and DDD estimation suggests a higher rates of resolution and lower filing rates among high-MRPK plaintiff firms.

Lastly, I estimate [Equation 3](#) using legal case data involving banks, separated by banks as plaintiff or as defendant. I find that better courts are less likely to resolve cases against banks (when banks are defendants, see [Table 8](#)). While the results are noisy, the coefficients

¹⁴In the event study by case-type, I find an increase in filing of cases relevant to debt recovery and bankruptcy process but I do not find any effects on other types of cases (using criminal cases as a placebo) across district judicial capacity before and after the reform.

¹⁵Since the outcome data is at the district court-firm-case-level and that the same firm could have cases in multiple courts, I can identify the coefficient on high judge-population districts interacted with post-reform dummy variable (which would otherwise be collinear with district-year fixed effects).

when banks appear as plaintiffs suggest increased resolution of cases per bank.

I rationalize these results as follows: high-default-risk firms are more likely to default on credit contracts, which results in cases being filed *against* such firms (who then appear as defendants). The results suggest that district courts with better capacity are more likely to resolve such cases when high-default-risk firms are defending, likely enforcing contracts. In contrast, when such firms file counter-suits as plaintiff (for example, to contest a previous judicial order), better capacity courts do not appear to take any action. Since courts differentially respond based on the specific role of banks/high-risk firms in litigation, the findings suggest that better courts support banks in their debt recovery strategy.

9 Conclusion

This paper provides evidence in support of the complementary role local courts play in enforcing creditor rights in the context of a national-level bankruptcy reform in India. Using a triple-difference research design, I show that this complementarity brings in credit disciplining but falls short of improving overall allocative efficiency. I find that these results are suggestively explained by local courts supporting banks in litigations through *executing* judicial orders against debt defaulters.

The results on credit allocation likely imply continuing information frictions where, perhaps lenders and institutions aren't fully able to discern high capital efficiency firms to increase lending to them. This is evident in the weak correlation between observed firm-level default risk (credit ratings) and capital efficiency (MRPK) indicators.

As the law takes full effect over time, the long run and general equilibrium effects may be different. These could have ramifications on what types of firms are liquidated and what types are restructured through re-investment, and therefore left for further research as and when such data becomes available. Furthermore, what this paper examines is capacity measured in terms of judge-to-population ratios, which is one specific aspect of judicial capacity. Other investments in the local enforcement environment, such as better triage through alternate dispute resolution and conciliation for enforcing any liquidation or asset transfers may have different impact, which remain as open questions for future research.

References

- Acemoglu, Daron and Simon Johnson**, “Unbundling Institutions,” *Journal of Political Economy*, 2005, *113* (5), 949–995. Publisher: The University of Chicago Press.
- Alok, Shashwat, Ritam Chaurey, and Vasudha Nukala**, “Creditors’ Rights, Threat of Liquidation, and the Labor and Capital Choices of Firms,” *The Journal of Law and Economics*, 2022, *65* (4), 687–714.
- Amirapu, Amrit**, “Justice Delayed Is Growth Denied: The Effect of Slow Courts on Relationship-Specific Industries in India,” *Economic Development and Cultural Change*, October 2021, *70* (1), 415–451. Publisher: The University of Chicago Press.
- Ash, Elliott, Sam Asher, Aditi Bhowmick, Sandeep Bhupatiraju, Daniel Chen, Tanaya Devi, Christoph Goessmann, Paul Novosad, and Bilal Siddiqi**, “In-Group Bias in the Indian Judiciary: Evidence from 5 Million Criminal Cases,” *The Review of Economics and Statistics*, 2025.
- Banerjee, Abhijit V. and Esther Duflo**, “Do Firms Want to Borrow More? Testing Credit Constraints Using a Directed Lending Program,” *The Review of Economic Studies*, April 2014, *81* (2), 572–607.
- Bau, Natalie and Adrien Matray**, “Misallocation and Capital Market Integration: Evidence From India,” *Econometrica*, 2023, *91* (1), 67–106.
- Besley, Timothy and Torsten Persson**, “The Origins of State Capacity: Property Rights, Taxation, and Politics,” *American Economic Review*, September 2009, *99* (4), 1218–44.
- Boehm, Johannes and Ezra Oberfield**, “Misallocation in the Market for Inputs: Enforcement and the Organization of Production*,” *The Quarterly Journal of Economics*, 06 2020, *135* (4), 2007–2058.
- Breza, Emily and Andres Liberman**, “Financial Contracting and Organizational Form: Evidence from the Regulation of Trade Credit,” *The Journal of Finance*, 2017, *72* (1), 291–324.
- **and Cynthia Kinnan**, “Measuring the Equilibrium Impacts of Credit: Evidence from the Indian Microfinance Crisis*,” *The Quarterly Journal of Economics*, 05 2021, *136* (3), 1447–1497.

- Buera, Francisco J., Joseph P. Kaboski, and Yongseok Shin**, “Finance and Development: A Tale of Two Sectors,” *American Economic Review*, August 2011, *101* (5), 1964–2002.
- Cantoni, Davide and Noam Yuchtman**, “Medieval Universities, Legal Institutions, and the Commercial Revolution *,” *The Quarterly Journal of Economics*, May 2014, *129* (2), 823–887.
- Chalfin, Aaron and Justin McCrary**, “Are U.S. Cities Underpoliced? Theory and Evidence,” *The Review of Economics and Statistics*, March 2018, *100* (1), 167–186.
- Chemin, Matthieu**, “Does Court Speed Shape Economic Activity? Evidence from a Court Reform in India,” *The Journal of Law, Economics, and Organization*, 11 2010, *28* (3), 460–485.
- , “Judicial Efficiency and Firm Productivity: Evidence from a World Database of Judicial Reforms,” *The Review of Economics and Statistics*, March 2020, *102* (1), 49–64.
- Chopra, Yakshup, Krishnamurthy Subramanian, and Prasanna L Tantri**, “Bank Cleanups, Capitalization, and Lending: Evidence from India,” *The Review of Financial Studies*, 10 2020, *34* (9), 4132–4176.
- Closset, Frédéric, Christoph Großmann, Christoph Kaserer, and Daniel Urban**, “Corporate restructuring and creditor power: Evidence from European insolvency law reforms,” *Journal of Banking & Finance*, April 2023, *149*, 106756.
- Cole, Shawn, Martin Kanz, and Leora Klapper**, “Incentivizing Calculated Risk-Taking: Evidence from an Experiment with Commercial Bank Loan Officers,” *The Journal of Finance*, 2015, *70* (2), 537–575. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/jofi.12233>.
- Dahiya, Sandeep, Kose John, Manju Puri, and Gabriel Ramírez**, “Debtor-in-possession financing and bankruptcy resolution: Empirical evidence,” *Journal of Financial Economics*, 2003, *69* (1), 259–280. Tuck Symposium on Corporate Governance.
- Demetriades, Panicos O. and Kul B. Luintel**, “Financial Development, Economic Growth and Banking Sector Controls: Evidence from India,” *The Economic Journal*, March 1996, *106* (435), 359–374. Publisher: Oxford Academic.
- Dhaliwal, Iqbal and Rema Hanna**, “The devil is in the details: The successes and limitations of bureaucratic reform in India,” *Journal of Development Economics*, January 2017, *124*, 1–21.

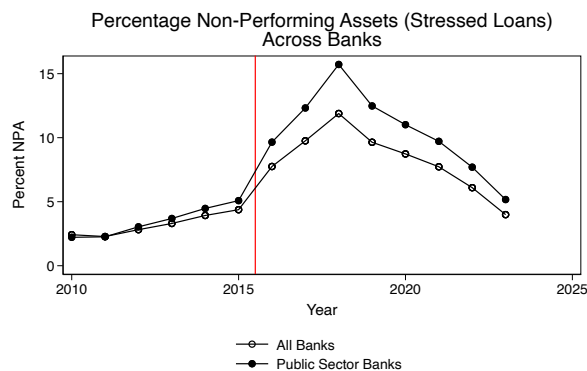
- Djankov, S, C Mcliesh, and A Shleifer**, “Private credit in 129 countries,” *Journal of Financial Economics*, May 2007, *84* (2), 299–329.
- Faria-e-Castro, Miguel, Pascal Paul, and Juan M. Sánchez**, “Evergreening,” *Journal of Financial Economics*, 2024, *153*, 103778.
- Fisman, Raymond, Daniel Paravisini, and Vikrant Vig**, “Cultural Proximity and Loan Outcomes,” *American Economic Review*, February 2017, *107* (2), 457–492.
- Giné, Xavier and Inessa Love**, “Do Reorganization Costs Matter for Efficiency? Evidence from a Bankruptcy Reform in Colombia,” *The Journal of Law and Economics*, November 2010, *53* (4), 833–864. Publisher: The University of Chicago Press.
- **and Martin Kanz**, “The Economic Effects of a Borrower Bailout: Evidence from an Emerging Market,” *The Review of Financial Studies*, May 2018, *31* (5), 1752–1783. Publisher: Oxford Academic.
- Glaeser, Edward L. and Andrei Shleifer**, “The Rise of the Regulatory State,” *Journal of Economic Literature*, June 2003, *41* (2), 401–425.
- Gormley, Todd, Nandini Gupta, and Anand Jha**, “Quiet Life No More? Corporate Bankruptcy and Bank Competition,” *The Journal of Financial and Quantitative Analysis*, 2018, *53* (2), 581–611. Publisher: [Cambridge University Press, University of Washington School of Business Administration].
- Hsieh, Chang-Tai and Peter J. Klenow**, “Misallocation and Manufacturing TFP in China and India*,” *The Quarterly Journal of Economics*, 11 2009, *124* (4), 1403–1448.
- India Justice Report** , “India Justice Report: Ranking States on the Capacity of Police, Judiciary, Prisons, and Legal Aid,” Technical Report 2025.
- Jack, William, Michael Kremer, Joost de Laat, and Tavneet Suri**, “Credit Access, Selection, and Incentives in a Market for Asset-Collateralized Loans: Evidence From Kenya,” *The Review of Economic Studies*, 03 2023, *90* (6), 3153–3185.
- Khwaja, Asim Ijaz and Atif Mian**, “Tracing the Impact of Bank Liquidity Shocks: Evidence from an Emerging Market,” *American Economic Review*, September 2008, *98* (4), 1413–1442.
- Kulkarni, Nirupama, S. K. Ritadhi, Siddharth Vij, and Katherine Waldock**, “Unearthing Zombies,” *Management Science*, January 2025. Publisher: INFORMS.

- Li, Bo and Jacopo Ponticelli**, “Going Bankrupt in China,” *Review of Finance*, May 2022, 26 (3), 449–486.
- Link, Sebastian, Manuel Menkhoff, Andreas Peichl, and Paul Schüle**, “Downward Revision of Investment Decisions after Corporate Tax Hikes,” *American Economic Journal: Economic Policy*, November 2024, 16 (4), 194–222.
- Müller, Karsten**, “Busy bankruptcy courts and the cost of credit,” *Journal of Financial Economics*, 2022, 143 (2), 824–845.
- **and Emil Verner**, “Credit Allocation and Macroeconomic Fluctuations,” *The Review of Economic Studies*, 12 2023, 91 (6), 3645–3676.
- Naaraayanan, S Lakshmi and Daniel Wolfenzon**, “Business Group Spillovers,” *The Review of Financial Studies*, January 2024, 37 (1), 231–264.
- Needleman, Jack, Peter Buerhaus, Soeren Mattke, Maureen Stewart, and Katya Zelevinsky**, “Nurse-Staffing Levels and the Quality of Care in Hospitals,” *New England Journal of Medicine*, May 2002, 346 (22), 1715–1722. Publisher: Massachusetts Medical Society _eprint: <https://www.nejm.org/doi/pdf/10.1056/NEJMsa012247>.
- Ortiz-Villavicencio, Marcelo and Pedro H. C. Sant’Anna**, “Better Understanding Triple Differences Estimators,” May 2025. arXiv:2505.09942 [econ].
- Ponticelli, Jacopo and Leonardo S. Alencar**, “Court Enforcement, Bank Loans, and Firm Investment: Evidence from a Bankruptcy Reform in Brazil,” *The Quarterly Journal of Economics*, August 2016, 131 (3), 1365–1413.
- Porta, Rafael La, Florencio Lopez-de-Silanes, Andrei Shleifer, and Robert W. Vishny**, “Law and Finance,” *Journal of Political Economy*, December 1998, 106 (6), 1113–1155.
- Rajan, Raghuram G. and Luigi Zingales**, “Financial Dependence and Growth,” *The American Economic Review*, 1998, 88 (3), 559–586.
- Rao, Manaswini**, “Front-line Courts As State Capacity: Evidence From India,” *Working Paper*, 2025.
- Vig, Vikrant**, “Access to Collateral and Corporate Debt Structure: Evidence from a Natural Experiment,” *The Journal of Finance*, 2013, 68 (3), 881–928.

- Visaria, Sujata**, “Legal reform and loan repayment: The microeconomic impact of debt recovery tribunals in India,” *American Economic Journal: Applied Economics*, 2009, *1* (3), 59–81.
- von Lilienfeld-Toal, Ulf, Dilip Mookherjee, and Sujata Visaria**, “THE DISTRIBUTIVE IMPACT OF REFORMS IN CREDIT ENFORCEMENT: EVIDENCE FROM INDIAN DEBT RECOVERY TRIBUNALS,” *Econometrica*, 2012, *80* (2), 497–558.
- von Lilienfeld-Toal, Ulf and Dilip Mookherjee**, “A General Equilibrium Analysis of Personal Bankruptcy Law,” *Economica*, 2016, *83* (329), 31–58.
- Wittrup, Jesper**, “Explaining Cross-Country Differences in the Number of Judges per Capita | International Journal for Court Administration,” *International Journal for Court Administration*, June 2024.
- Òscar Jordà, Martin Kornejew, Moritz Schularick, and Alan M Taylor**, “Zombies at Large? Corporate Debt Overhang and the Macroeconomy,” *The Review of Financial Studies*, October 2022, *35* (10), 4561–4586.

10 Figures

Figure 1: NPA Across Banks in India



Notes: The figure above graphs the percentage of total lending declared as a non-performing asset (NPA) aggregated across all commercial banks in India and by public sector banks, as reported by the central bank, Reserve Bank of India. The red vertical line indicates when the bankruptcy bill was first tabled in the national parliament. I present the timeline of the reform below.

Figure 2: Timeline of Bankruptcy Reform

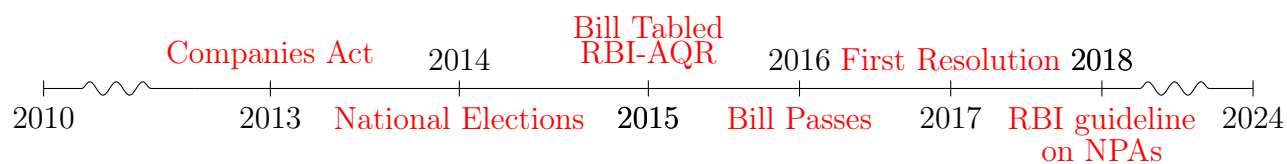
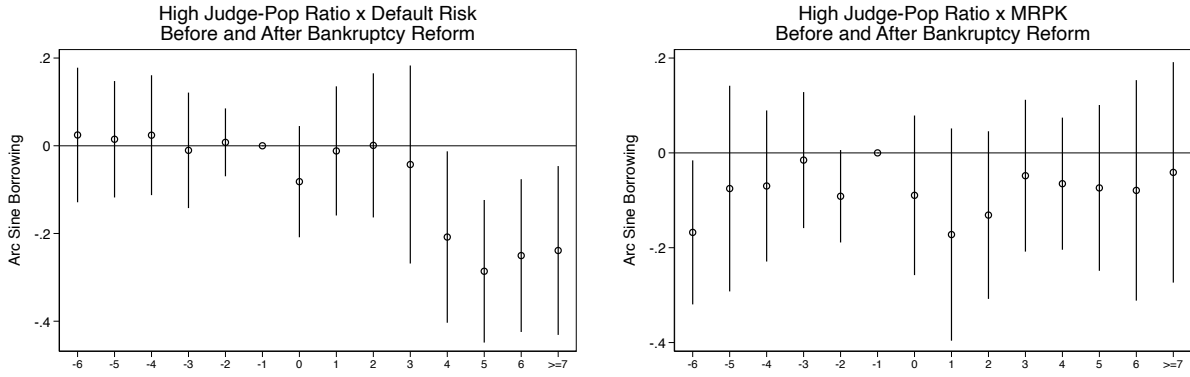
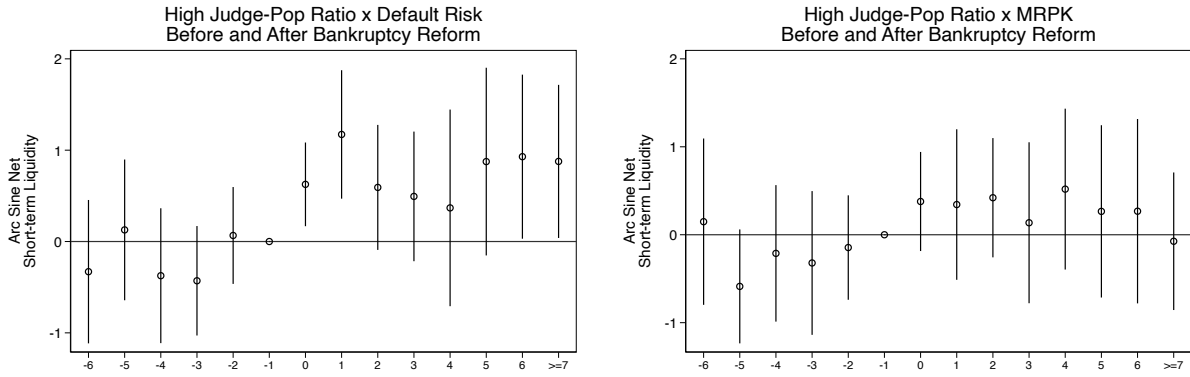


Figure 3: Borrowing and Liquidity Outcomes

Panel A: Borrowing

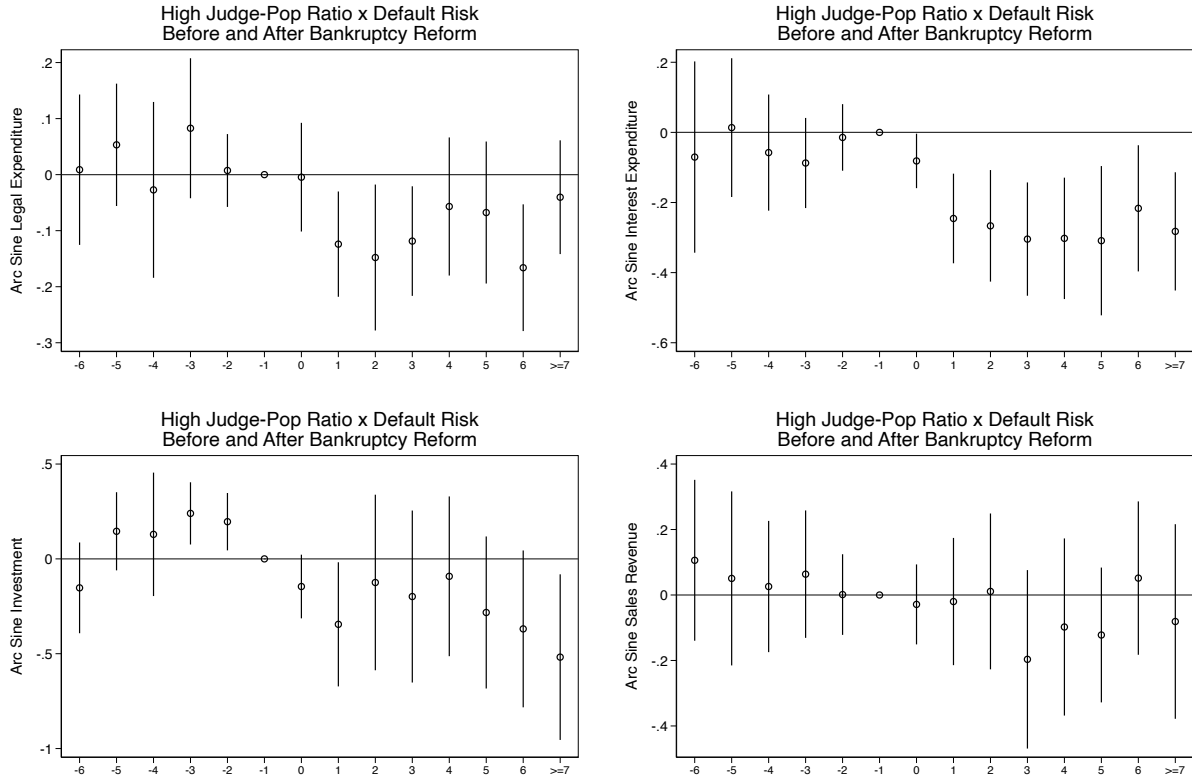


Panel B: Liquidity



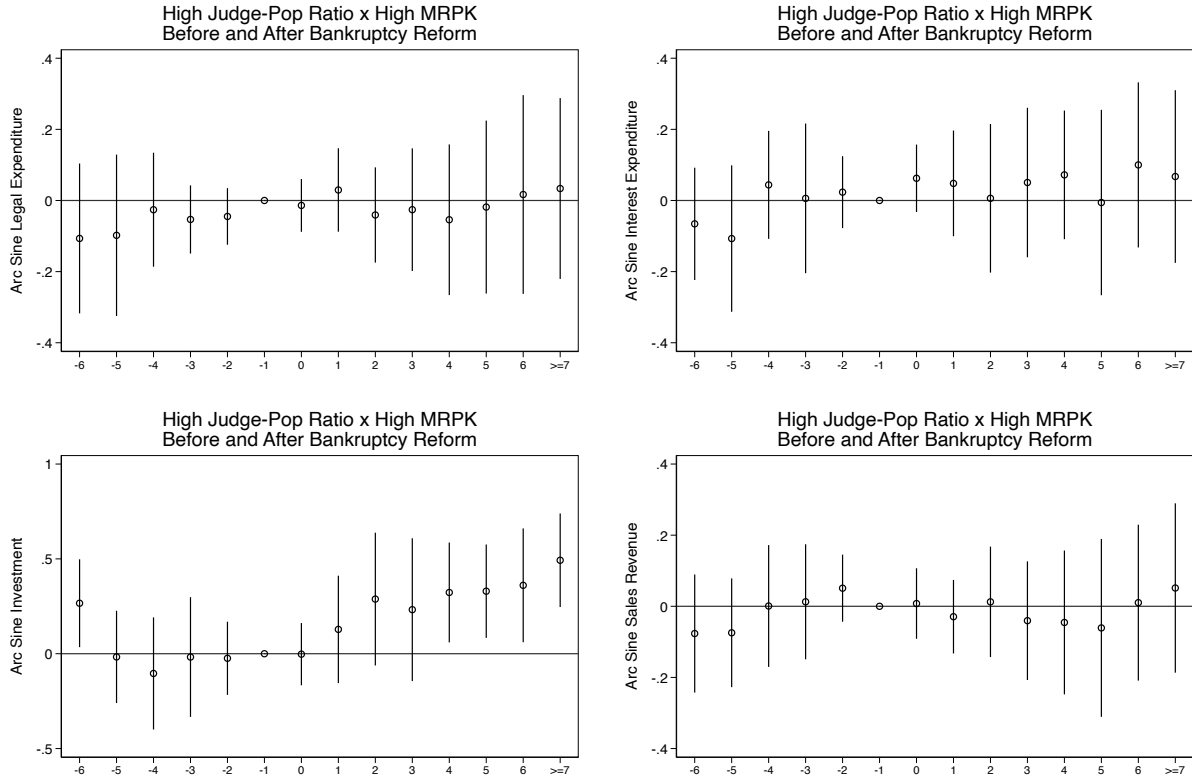
Notes: Panels A and B above present the estimates of event study implementation of [Equation 1](#) using borrowing (Panel A) and short-term liquidity (Panel B) outcomes as dependent variables for high-default-risk and high-MRPK interactions, respectively. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. High judge to population ratio districts are those with the ratio above the sample average. High default risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) before 2010. High-MRPK firms are those with estimated pre-2010 MRPK above median of the firm-level sample. Error bars present 95% confidence interval. Standard errors are clustered by district. The DDD estimates are reported in [Table 2](#).

Figure 4: Economic Outcomes: By Baseline Default Risk



Notes: The figures above present the triple interaction estimates of event study implementation of [Equation 1](#) using financial and economic outcomes as dependent variables for firms classified as high-default risk in high judge-population ratio districts. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. High judge to population ratio districts are those with the ratio above the sample average. High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) in the period prior to the bankruptcy reform. Error bars present 95% confidence interval. Standard errors are clustered by district. The DDD estimates are reported in [Table 4](#).

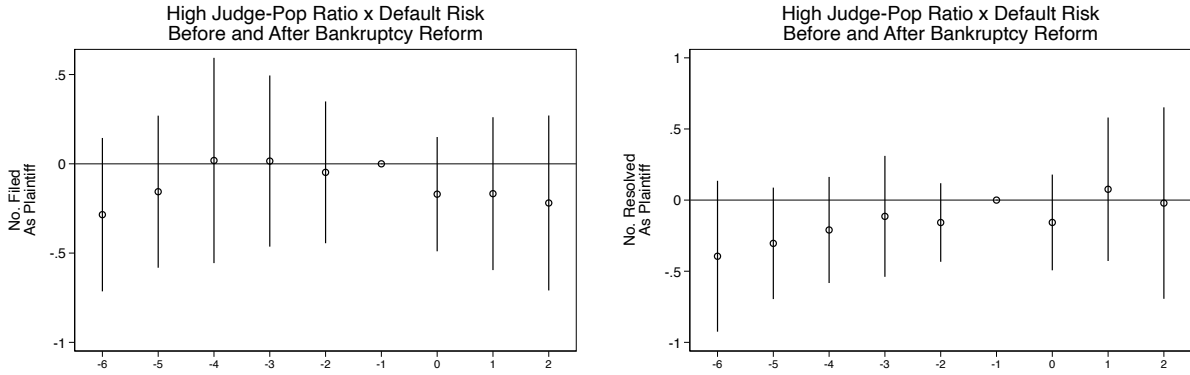
Figure 5: Economic Outcomes: By Baseline MRPK



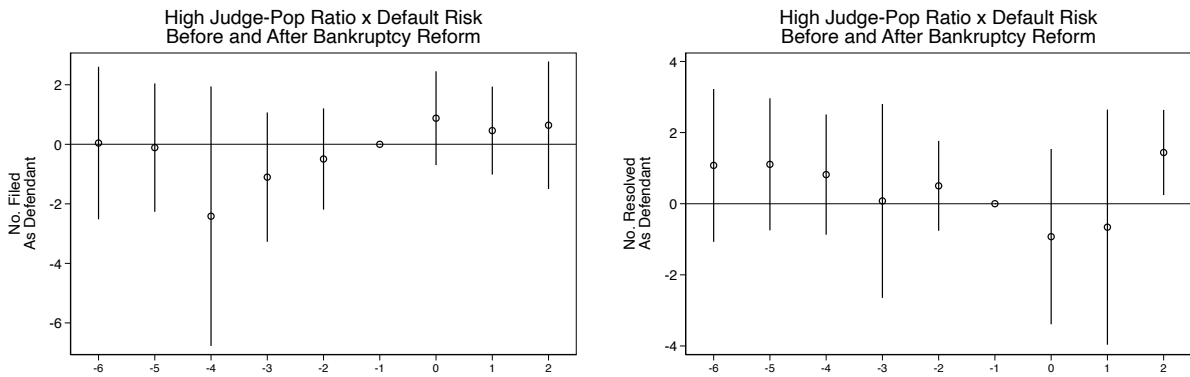
Notes: The figures above present the triple interaction estimates of event study implementation of Equation 1 using financial and economic outcomes as dependent variables for firms classified as high-MRPK in high judge-population ratio districts. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. High judge to population ratio districts are those with the ratio above the sample average. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Error bars present 95% confidence interval. Standard errors are clustered by district. The DDD estimates are reported in Table 5.

Figure 6: Litigation in Courts by Firms: By Baseline Default-Risk

Panel A: As Plaintiff

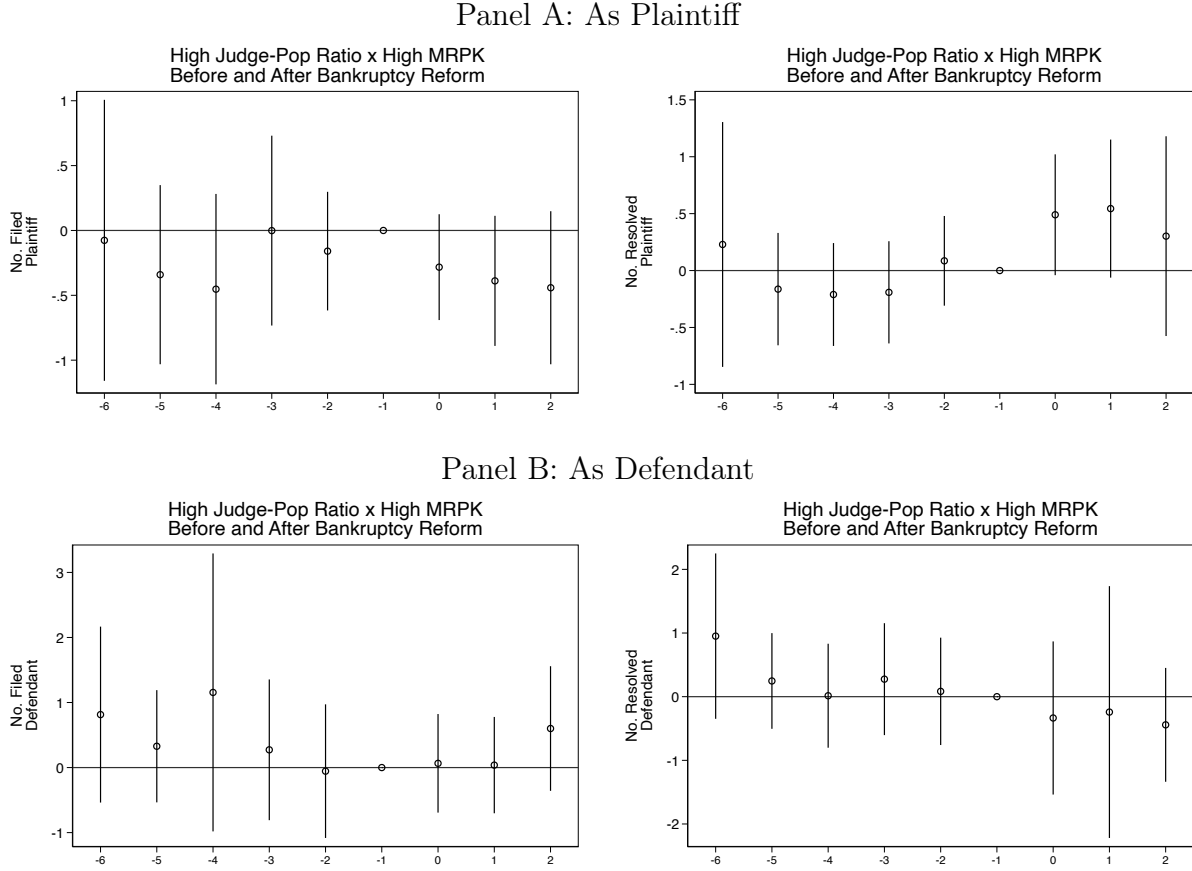


Panel B: As Defendant



Notes: The figures above present the triple interaction estimates of event study implementation of [Equation 2](#) using legal case-level outcomes in the corresponding district court as dependent variables for firms classified as high default risk in high judge-population ratio districts. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. Note that the judicial case-level data is only available until 2018, so there are fewer post reform time periods compared to the analysis using firm-level outcomes. High judge to population ratio districts are those with the ratio above the sample average. High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) in the period prior to the bankruptcy reform. Panel A presents the estimates on case-level outcomes when high-risk firms appear as plaintiff in a case. That is, when such firms initiate the suit in courts. Panel B presents the estimates when such firms appear as defendants in a case, i.e., when someone else files a suit against them. Error bars present 95% confidence interval. Standard errors are clustered by district. The DDD estimates are reported in [Table 6](#).

Figure 7: Litigation in Courts by Firms: By Baseline MRPK



Notes: The figures above present the triple interaction estimates of event study implementation of [Equation 2](#) using legal case-level outcomes in the corresponding district court as dependent variables for firms classified as high-MRPK in high judge-population ratio districts. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. Note that the judicial case-level data is only available until 2018, so there are fewer post reform time periods compared to the analysis using firm-level outcomes. High judge to population ratio districts are those with the ratio above the sample average. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Panel A presents the estimates on case-level outcomes when high-MRPK firms appear as plaintiff in a case. That is, when such firms initiate the suit in courts. Panel B presents the estimates when such firms appear as defendants in a case, i.e., when someone else files a suit against them. Error bars present 95% confidence interval. Standard errors are clustered by district. The DDD estimates are reported in [Table 7](#).

11 Tables

Table 1: Summary Statistics

	(1) No. Firms	(2) Mean	(3) SD
Judge to Population Ratio (per million)	3582	51.74	49.60
Estimated MRPK	3381	9.09	41.76
Credit Rating (1 lowest-8 highest)	3582	5.78	1.78
Public Ltd.	3582	0.80	0.40
Age (years)	3580	40.82	19.36
Share Manufacturing	3582	0.59	0.49
Share Construction	3582	0.11	0.31
Share Trade/Retail	3582	0.19	0.39
Share Services	3582	0.02	0.15
Sales Rev (Million INR)	3411	4605.24	25855.26
Total Assets (Million INR)	3582	5961.70	31772.94

Notes: The main sample includes over 3000 formal, non-financial sector firms registered in districts with courts classified as high or low judicial capacities (judge to population ratio) that form the main population of interest for this study. Note that the judge to population ratio in this sample is different from the national average of 18.6, which includes districts with no registered formal sector firms. Age is calculated as the difference between current year and the year of firm's incorporation. The remaining 8% of sample firms belong to other non-financial sectors represented by industrial classification (NIC) codes that could not be neatly binned into the sector categories above. The average sales revenue and asset value are computed for period prior to the reform.

Table 2: Bankruptcy Reform and District Judicial Capacity: Firms' Borrowing and Liquidity

	(1) Arc Sine Borrowing	(2) Arc Sine Net Liquidity	(3) Arc Sine Borrowing	(4) Arc Sine Net Liquidity
High Jdg Pop Ratio x High Default Risk x Post	-0.156** (0.0678)	0.818*** (0.299)		
High Default Risk x Post	0.0856 (0.0675)	-0.608*** (0.216)		
High Jdg Pop Ratio x High MRPK x Post			-0.00876 (0.0532)	0.331 (0.261)
High MRPK x Post			0.0609 (0.0487)	-0.531** (0.225)
Observations	57840	57885	47970	48000
No. Districts	164	164	150	150
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	7.400	2.360	7.480	2.060
SD Dep Var	2.200	6.520	2.290	6.700
Adj R-Squared	0.801	0.624	0.797	0.616

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: Col 1-2 above present the estimates from Equation 1 using borrowing and short-term liquidity outcomes as dependent variables for high-default-risk firm interactions. Col 3-4 present the estimates for high-MRPK firm interactions. "Post" is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. High-judge-to-population ratio districts are those with the ratio above the sample average. High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) in the period prior to the bankruptcy reform. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Standard errors are clustered by district. The event study figures are in Figure 3.

Table 3: Firm Default Risk and MRPK

	(1) High MRPK	(2) High MRPK	(3) High MRPK
High Default Risk	-0.0918*** (0.0266)	-0.0404 (0.0265)	-0.0460 (0.0281)
Constant	0.482*** (0.00883)	0.474*** (0.00772)	0.478*** (0.00788)
Observations	3582	3431	3347
Fixed Effect	None	Industry	Industry, District
Adj R-Squared	0.00291	0.257	0.259

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table reports firm-level correlations between default risk and their high-MRPK classification using data from pre-reform period. High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) in the period prior to the bankruptcy reform. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Col 1 presents simple, bi-variate correlation. Cols 2-3 include location and industry fixed effects to account for risk or productivity factors driven either by location of registration or industry. Heteroskedasticity robust standard errors reported in parentheses.

Table 4: Bankruptcy Reform and District Judicial Capacity: Firms' Production Outcomes by Default Risk

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
High Jdg Pop Ratio x High Default Risk x Post	-0.112*** (0.0398)	-0.234*** (0.0538)	-0.365* (0.200)	-0.0982 (0.0742)
High Default Risk x Post	-0.0224 (0.0407)	0.0401 (0.0458)	0.0283 (0.0845)	-0.0337 (0.0593)
Observations	50190	57210	44700	57360
No. Districts	152	164	145	163
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.870	5.050	5.330	8.650
SD Dep Var	1.720	1.830	2.920	1.870
Adj R-Squared	0.906	0.805	0.878	0.871

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using legal and interest expenditures, long-run investment (maturity > 1 year), and sales revenue as dependent variables for high-default-risk firm interactions. "Post" is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. High-judge-to-population ratio districts are those with the ratio above the sample average. High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) in the period prior to the bankruptcy reform. Standard errors are clustered by district. The event study figures are in Figure 4.

Table 5: Bankruptcy Reform and District Judicial Capacity: Firms' Production Outcomes by MRPK

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
High Jdg Pop Ratio x High MRPK x Post	0.0458 (0.0618)	0.0560 (0.0607)	0.316*** (0.0707)	0.00498 (0.0495)
High MRPK x Post	-0.0398 (0.0278)	-0.0310 (0.0440)	-0.0464 (0.0713)	-0.132*** (0.0401)
Observations	41070	47400	36795	47520
No. Districts	136	150	130	149
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.910	5.140	5.540	8.690
SD Dep Var	1.760	1.890	2.950	1.940
Adj R-Squared	0.905	0.801	0.877	0.873

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from [Equation 1](#) using legal and interest expenditures, long-run investment (maturity > 1 year), and sales revenue as dependent variables for high-MRPK firm interactions. "Post" is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. High-judge-to-population ratio districts are those with the ratio above the sample average. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Standard errors are clustered by district. The event study figures are in [Figure 5](#).

Table 6: Firm Default Risk and Court Trials

	(1)	(2)	(3)	(4)	(5)	(6)
	No. Filed Defendant	No. Resolved Defendant	No. Pending Defendant	No. Filed Plaintiff	No. Resolved Plaintiff	No. Pending Plaintiff
High Jdg Pop Ratio x High Default Risk x Post	1.008 (0.614)	0.0107 (1.143)	2.855* (1.566)	-0.104 (0.174)	0.219 (0.238)	-0.285 (0.294)
High Jdg Pop Ratio x Post	-0.846 (0.551)	0.456 (0.816)	-2.551* (1.514)	-0.202 (0.350)	0.0467 (0.529)	-0.587 (0.622)
High Default Risk x Post	-0.334 (0.250)	-0.238 (0.351)	-0.356 (0.401)	0.0828 (0.137)	-0.237 (0.226)	0.248 (0.227)
Observations	17037	17037	17037	47340	47340	47340
No. Districts	146	146	146	234	234	234
Firm Fixed Effect	X	X	X	X	X	X
District-Year FE	X	X	X	X	X	X
Industry-Year FE	X	X	X	X	X	X
Mean Dep Var	0.530	0.140	0.390	0.170	0.0200	0.160
SD Dep Var	3.560	1.860	2.540	3.760	0.340	3.540
Adj R-Squared	0.0560	0.0000128	0.120	0.0847	0.0786	0.131

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 2 using legal case-level outcomes in the corresponding district court as dependent variables for firms classified as high default risk in high-judge-population-ratio districts. “Post” is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. Note that the judicial case-level data is only available until 2018, so there are fewer post reform time periods compared to the analysis using firm-level outcomes.

High-judge-to-population ratio districts are those with the ratio above the sample average.

High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody’s) in the period prior to the bankruptcy reform.

Col 1-3 present the estimates when such firms appear as defendants in a case, i.e., when someone else files a suit against them. Col 4-6 presents the estimates on case-level outcomes when these firms appear as plaintiff in a case, i.e, when such firms initiate lawsuits in courts. Since the outcome data is at the district court-firm-case-level and that the same firm could have cases in multiple courts, I can identify the coefficient on high judge-population districts interacted with post-reform dummy variable (which would otherwise be collinear with district-year fixed effects). Error bars present 95% confidence interval.

Standard errors are clustered by district. The event study estimates are reported in Figure 6.

Table 7: Firm MRPK and Court Trials

	(1)	(2)	(3)	(4)	(5)	(6)
	No. Filed Defendant	No. Resolved Defendant	No. Pending Defendant	No. Filed Plaintiff	No. Resolved Plaintiff	No. Pending Plaintiff
High Jdg Pop Ratio x High MRPK x Post	-0.0496 (0.279)	-0.519 (0.545)	0.0687 (0.764)	-0.228 (0.188)	0.389 (0.303)	-0.506 (0.315)
High Jdg Pop Ratio x Post	-1.051** (0.519)	0.666 (0.885)	-2.851** (1.425)	-0.282 (0.367)	-0.322 (0.537)	-0.642 (0.600)
High MRPK x Post	-0.875 (0.575)	0.662 (0.744)	-2.518* (1.280)	0.341 (0.302)	-0.214 (0.273)	0.625 (0.558)
Observations	15291	15291	15291	31761	31761	31761
No. Districts	144	144	144	213	213	213
Firm Fixed Effect	X	X	X	X	X	X
District-Year FE	X	X	X	X	X	X
Industry-Year FE	X	X	X	X	X	X
Mean Dep Var	0.440	0.0900	0.350	0.230	0.0200	0.210
SD Dep Var	3.100	0.890	2.530	4.520	0.390	4.260
Adj R-Squared	0.0692	-0.00140	0.129	0.0688	0.0630	0.110

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 2 using legal case-level outcomes in the corresponding district court as dependent variables for firms classified as high-MRPK in high judge-population ratio districts. “Post” is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. Note that the judicial case-level data is only available until 2018, so there are fewer post reform time periods compared to the analysis using firm-level outcomes. High-judge-to-population ratio districts are those with the ratio above the sample average. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Col 1-3 present the estimates when such firms appear as defendants in a case, i.e., when someone else files a suit against them. Col 4-6 presents the estimates on case-level outcomes when high-MRPK firms appear as plaintiff in a case. That is, when such firms initiate the suit in courts. Since the outcome data is at the district court-firm-case-level and that the same firm could have cases in multiple courts, I can identify the coefficient on high judge-population districts interacted with post-reform dummy variable (which would otherwise be collinear with district-year fixed effects). Error bars present 95% confidence interval. Standard errors are clustered by district. The event study estimates are reported in Figure 7.

Table 8: Banks and Court Trials

	(1) No. Filed Bank Defendant	(2) No. Resolved Bank Defendant	(3) No. Pending Bank Defendant	(4) No. Filed Bank Plaintiff	(5) No. Resolved Bank Plaintiff	(6) No. Pending Bank Plaintiff
High Jdg Pop Ratio x Post	-0.0382 (0.0253)	-0.124*** (0.0369)	0.0281 (0.0447)	0.0907 (0.102)	0.135 (0.222)	-0.125 (0.0821)
Observations	26073	26073	26073	28008	28008	28008
No. Districts	104	104	104	104	104	104
Bank Fixed Effect	X	X	X	X	X	X
District Fixed Effect	X	X	X	X	X	X
Mean Dep Var	1.560	0.250	1.310	1.930	0.290	1.640
SD Dep Var	9.240	2.860	7.380	17.27	5.910	13.52
Adj R-Squared	0.264	0.248	0.256	0.0529	0.0749	0.109

Standard errors in parentheses

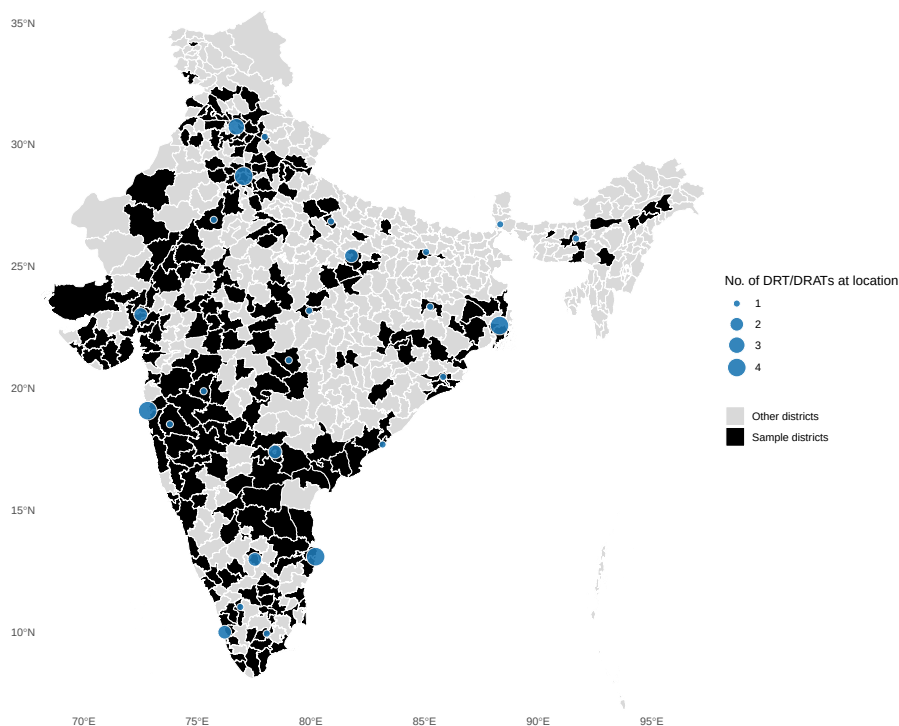
* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from [Equation 3](#) using legal case-level outcomes in the corresponding district court as dependent variables for cases involving banks. “Post” is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. Note that the judicial case-level data is only available until 2018, so there are fewer post reform time periods compared to the analysis using firm-level outcomes. High-judge-to-population ratio districts are those with the ratio above the sample average. Col 1-3 present the estimates when banks appear as defendants in a case, i.e., when someone else files a suit against them, such as to prevent liquidation. Col 4-6 presents the estimates on case-level outcomes when banks appear as plaintiff in a case, for example, when they need to enforce a bankruptcy order. Error bars present 95% confidence interval. Standard errors are clustered by district.

Online Appendix

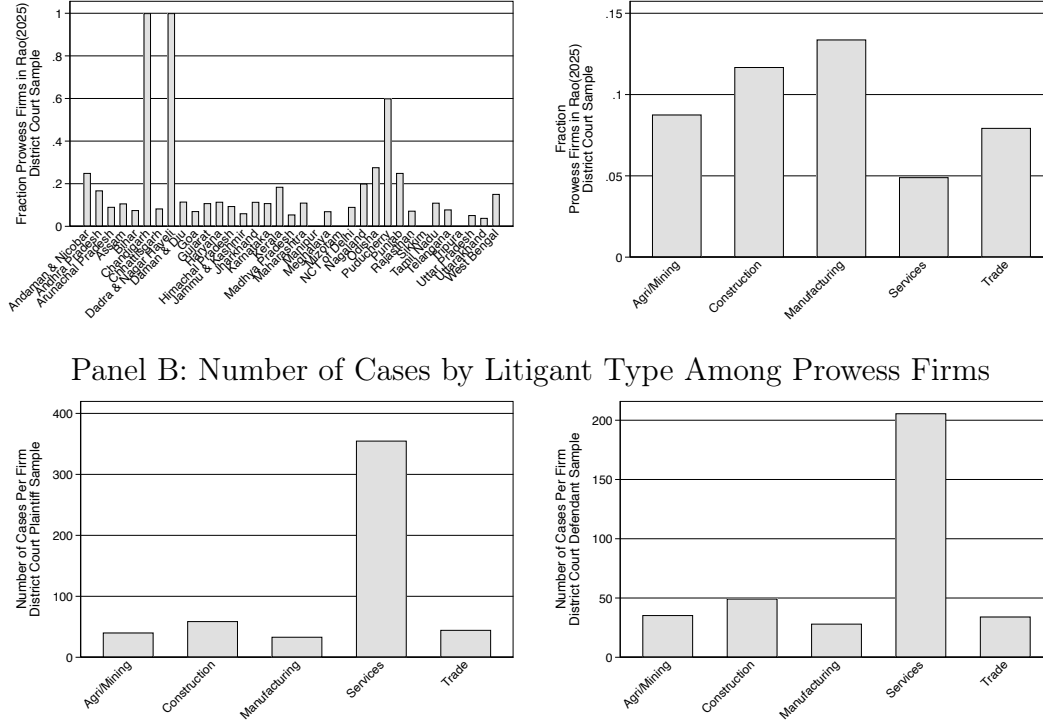
Additional figures and tables

Figure A.1: Sample Districts and DRT/NCLT Location



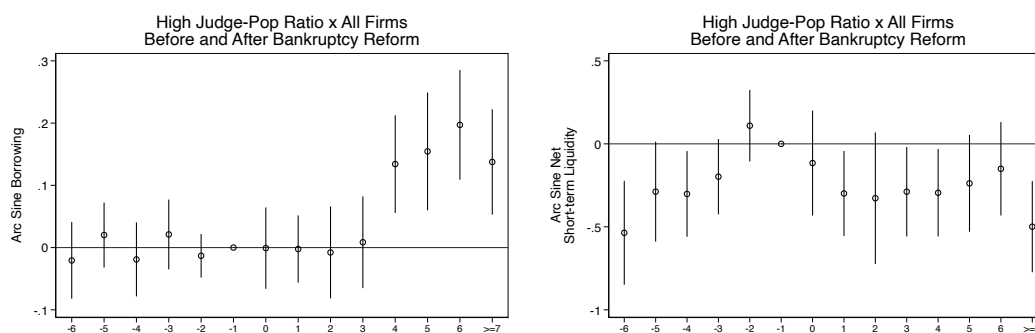
Notes: The above map shows the spatial distribution of district courts in the matched-firm sample and DRT locations. There are 243 unique district courts in the sample whereas DRTs are concentrated only in 25 locations, which overlap with the district court sample as well as the location of NCLT benches.

Figure A.2: Match Between Prowess Firms and Litigating Firms in District Courts
Panel A: Prowess Firm Coverage by State and Sector



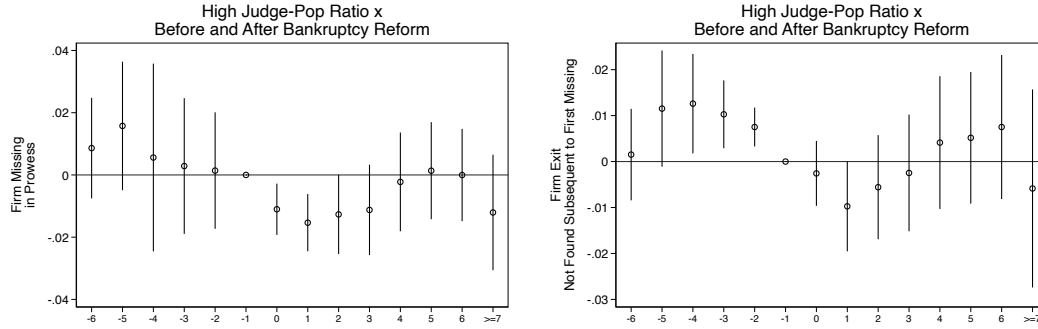
Notes: The above map shows the spatial and sectoral distribution of non-financial Prowess firms found in the district court case-level data from [Rao \(2025\)](#). Panel A shows the distribution of Prowess firms in the case-level data by the state of their registered office location and sector, respectively. This covers about 11% of Prowess firms on average. Panel B shows distribution of cases by Prowess firm sector (number of cases per firm) separately by their status as a plaintiff or defendant in the case-level data.

Figure A.3: Borrowing/Liquidity effects of bankruptcy reform on all firms by district judicial capacity

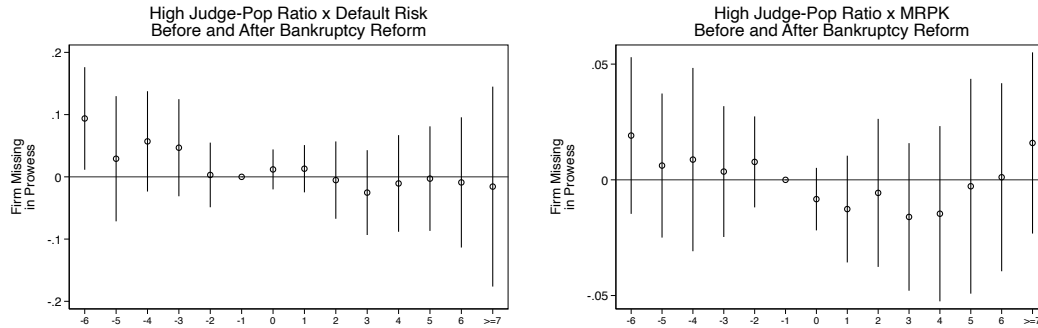


Notes: The figures report the event study estimates of bankruptcy reform on all firms by comparing those registered in high-judicial-capacity districts, measured using baseline judge-to-population ratio, with those registered in low capacity districts using borrowing and short-run liquidity as outcome variables. The error bars denote 95% confidence intervals and standard errors are clustered by district. These specifications are used to test the trends in borrowing and liquidity outcomes of firms in the period before the reforms.

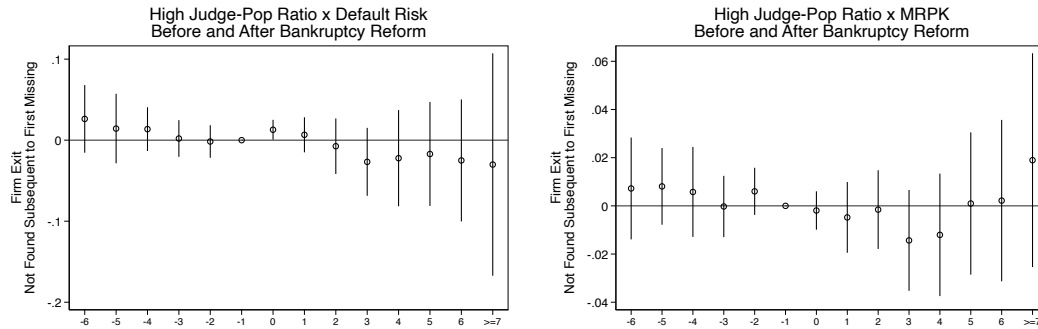
Figure A.4: Unobserved Firm Exits: Missing Financial Variables in Prowess Data Over Time
Panel A: DiD using All Firms



Panel B: Missing Reporting (Triple-D) By Default-Risk or MRPK

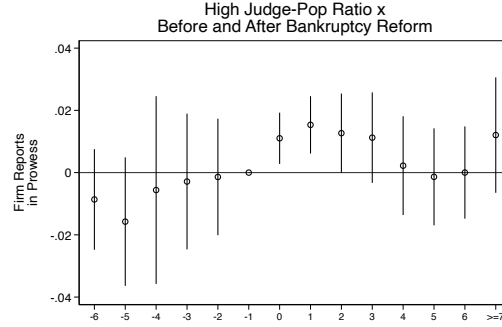


Panel C: Permanent Missing (Triple-D) By Default-Risk or MRPK

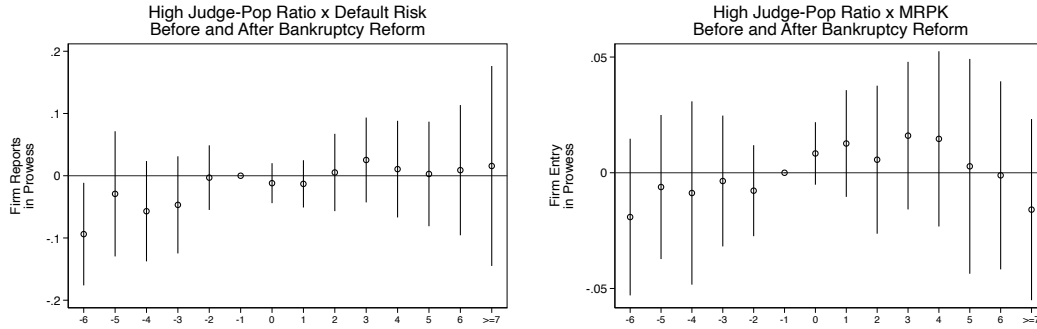


Notes: The figures depict the effect of the reform on firm disappearance in data using all Prowess firms in the sample districts. Formal firm exits through liquidations are unobserved in the data. I code firm disappearance in data along two margins using Prowess. First, I measure whether a firm does not report their financial data for any year in the sample time period 2010-2024. I call this “Firm Missing” to indicate years where I don’t observe the corresponding financial data for the firm. Second, I measure whether a firm completely stops reporting data after the first time they stop reporting. This indicates whether the firm exits the Prowess dataset. Panel A presents event study using these measures as the dependent variable in DiD design comparing the outcome between high judicial capacity districts with those in low capacity districts. Panel B and Panel C present the triple difference event study results by firms classified as high default risk or high MRPK, respectively, and the underlying judicial capacities of the district. All specifications include firm fixed effects, and industry-year fixed effects and the triple difference specifications additionally use district-year fixed effects. The error bars denote 95% confidence intervals and standard errors are clustered by district.

Figure A.5: Selection into Reporting: Financial Variables in Prowess Data Over Time
Panel A: DiD using All Firms

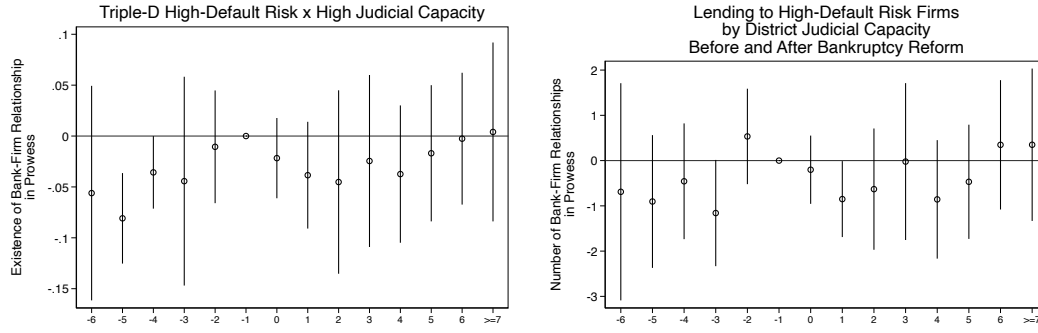


Panel B: Triple-D By Default-Risk or MRPK

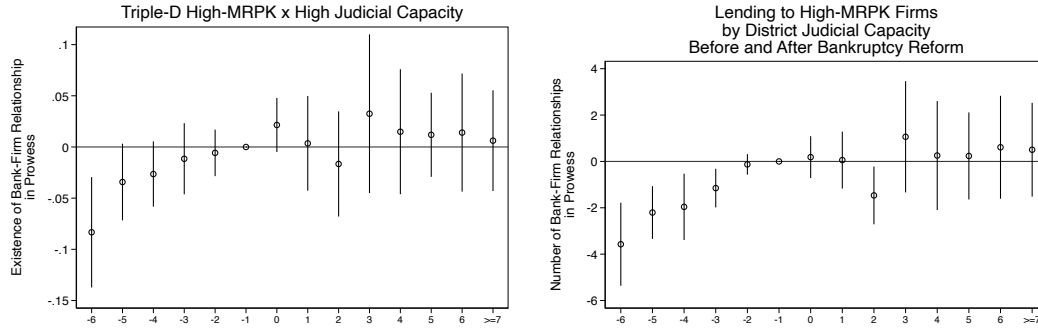


Notes: The figures depict the effect of the reform on firms' reporting of financial variables (non-missingness) in data using all Prowess firms in the sample districts. I measure whether a firm reports their financial data for each year in the sample time period 2010-2024. I call this "Firm Reports" to indicate years where I do observe the corresponding (non-missing) financial data for the firm. Panel A presents event study using this measure as the dependent variable in DiD design comparing the outcome between high judicial capacity districts with those in low capacity districts. Panel B presents the triple difference event study results by firms classified as high default risk or high MRPK, respectively, and the underlying judicial capacities of the district. All specifications include firm fixed effects, and industry-year fixed effects and the triple difference specifications additionally use district-year fixed effects. The error bars denote 95% confidence intervals and standard errors are clustered by district.

Figure A.6: Firm-Bank Relationship
Panel A: Triple-D By Default-Risk

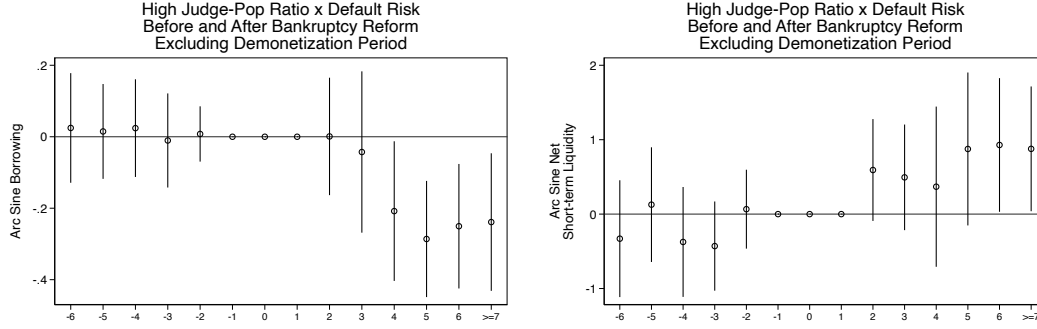


Panel B: Triple-D By MRPK

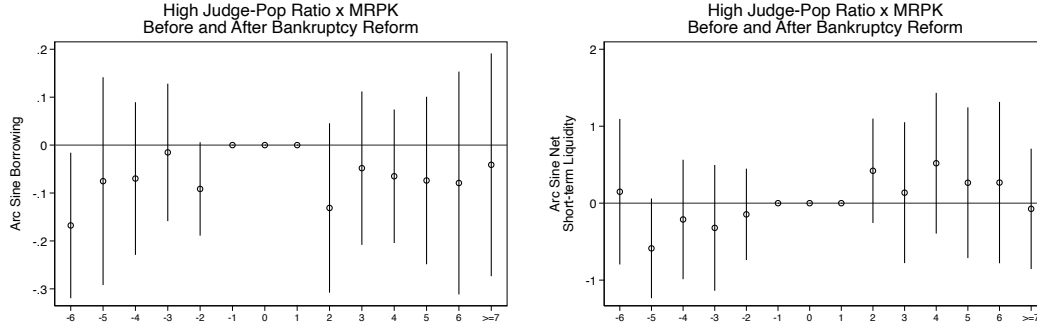


Notes: The figures depict the effect of the reform on firm-bank relationship observed in the “banker” table corresponding to the Prowess firm sample. There are two relevant margins of this relationship: first, whether there exists a relationship between a firm and a bank and second, total number of banking relationships by a firm. There are 658 unique banks with at least one relationship with the sample firm. The banks span commercial banks as well as regional banks and credit unions. The dependent variable “Existence of Bank-Firm Relationship” is a dummy variable that takes value 1 when I observe a relationship between a specific bank and a sample firm for a given year and 0 otherwise. The variable “Number of Bank-Firm Relationships” is constructed by aggregating the existence indicator across all banks for a specific firm for a given calendar year. The data includes a balanced panel of all sample firms over the study period 2010-2024. Additionally, the existence indicator also includes the specific banking partner for the firm. For regressions using the existence variable, I also include bank fixed effects in addition to firm and district-year fixed effects. Regressions using the aggregate variable includes only firm and district-year fixed effects. Panels A and B present the triple difference event study results by firms classified as high default risk or high MRPK, respectively, and the underlying judicial capacities of the district. The error bars denote 95% confidence intervals and standard errors are clustered by district.

Figure A.7: Robustness to other concurrent shocks: Demonetization, GST
Panel A: High Default-Risk Firms

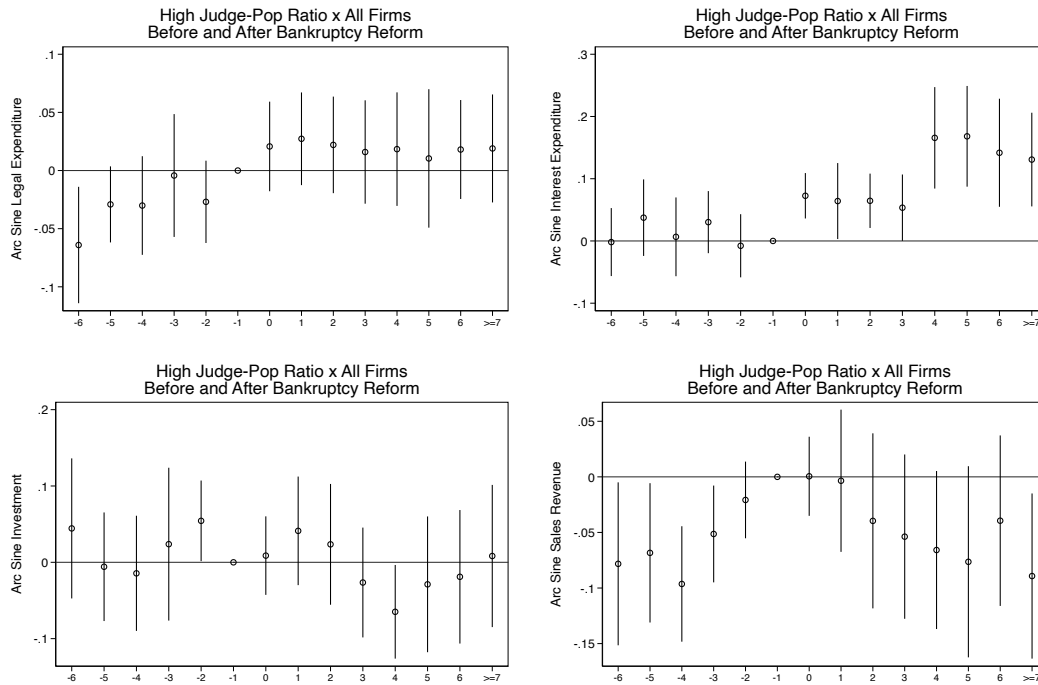


Panel B: High MRPK Firms



Notes: Panels A and B above present the estimates of event study implementation of [Equation 1](#) using borrowing (Panel A) and short-term liquidity (Panel B) outcomes as dependent variables for high-default-risk and high-MRPK interactions, respectively. These figures exclude the years 2016 and 2017 with many, concurrent national policy shocks such as demonetization in 2016 and the roll-out of GST in 2017. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. High judge to population ratio districts are those with the ratio above the sample average. High default risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) before 2010. High-MRPK firms are those with estimated pre-2010 MRPK above median of the firm-level sample. Error bars present 95% confidence interval. Standard errors are clustered by district.

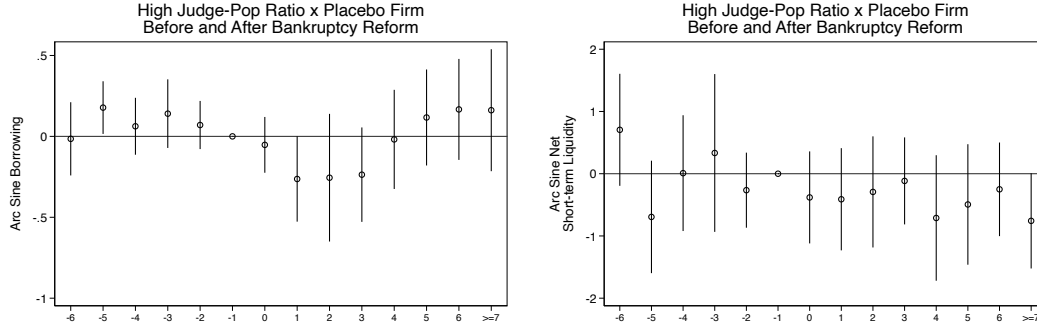
Figure A.8: Balance sheet effects of bankruptcy reform on all firms by district judicial capacity



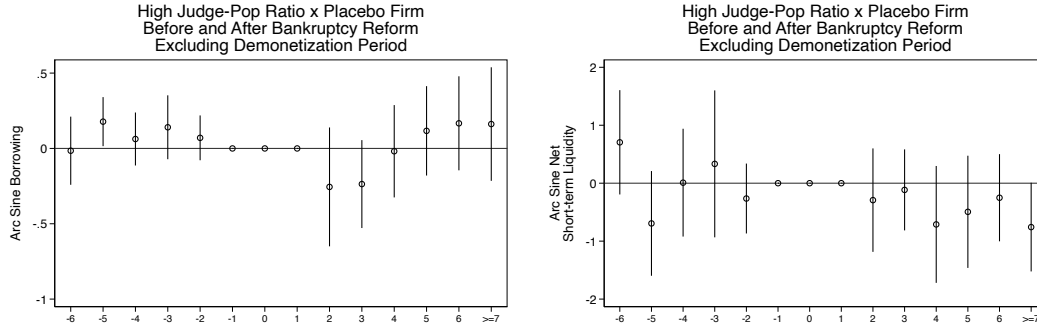
Notes: The figures report the event study estimates of bankruptcy reform on all firms by comparing those registered in high-judicial-capacity districts, measured using baseline judge-to-population ratio, with those registered in low capacity districts using balance sheet outcome variables including legal and interest expenditures, long run investments and sales revenue. The error bars denote 95% confidence intervals and standard errors are clustered by district. These specifications are used to test the trends in outcomes of firms in the period before the reforms.

Figure A.9: Placebo Firm Analysis

Panel A: Borrowing and Liquidity Effects Among Placebo Firms

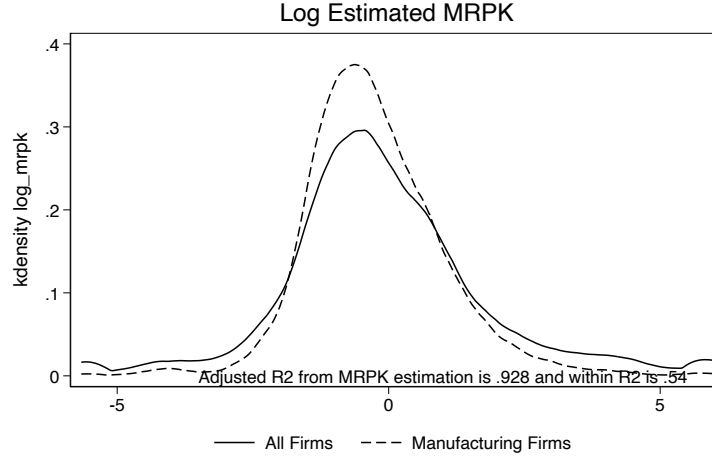


Panel B: Excluding Demonetization/GST Years

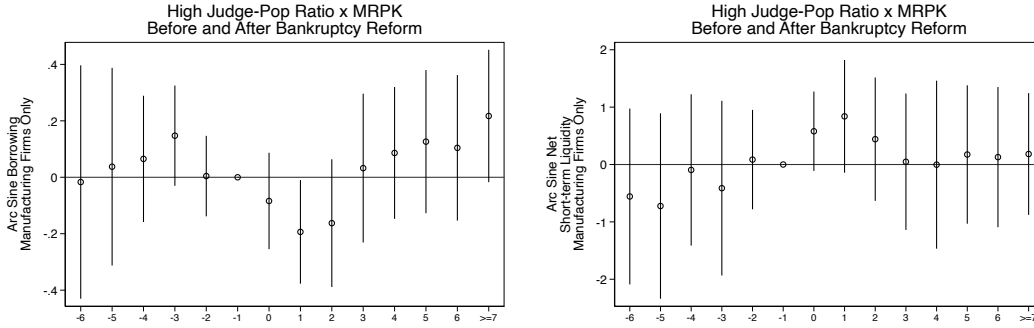


Notes: Panel A presents the estimates of event study implementation of [Equation 1](#) using borrowing and short-term liquidity outcomes as dependent variables for placebo-firm district judicial capacity interactions, respectively. Placebo firms are those with perfect credit ratings in the 2005-2010 time period. Credit allocation to such firms should not theoretically be affected by institutional quality or lenders' ever-greening motives incentives. Figures in Panel B additionally exclude the years 2016 and 2017 with many, concurrent national policy shocks such as demonetization in 2016 and the roll-out of GST in 2017. In all figures, the reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. High judge to population ratio districts are those with the ratio above the sample average. Error bars present 95% confidence interval. Standard errors are clustered by district.

Figure A.10: MRPK Estimation and Accounting for Measurement Error
 Panel A: Estimated MRPK Distribution



Panel B: Borrowing and Liquidity Effects by MRPK in Manufacturing

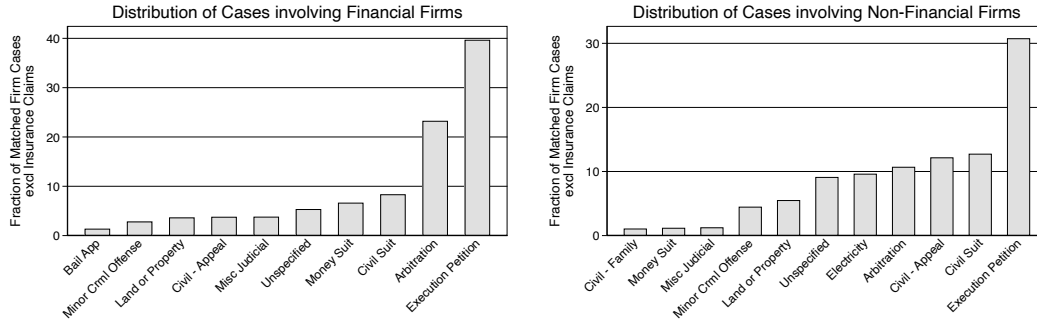


Notes: Panel A presents the kernel density distribution of Log estimated MRPK using the following procedure:

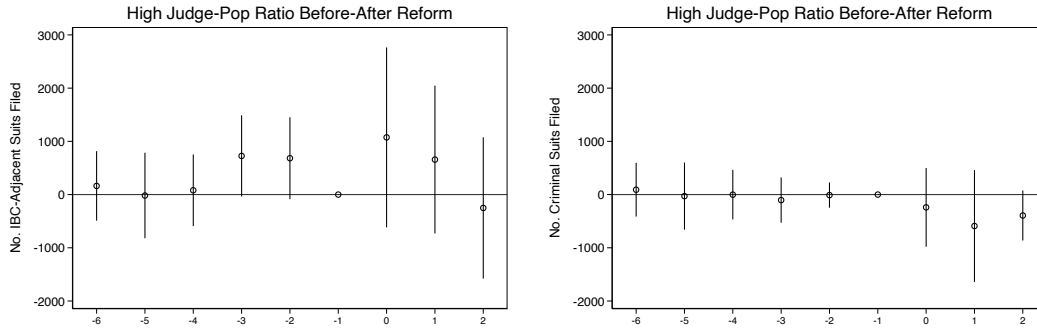
$$y_{it} = \beta_k K_{it} + \beta_l L_{it} + \phi_i + \phi_t + \varepsilon_{it}$$

where y_{it} is log sales revenue, K_{it} is the log value of plant and machinery, L_{it} is log wagebill. Data from before 2010 is used for estimating MRPK as: $\widehat{MRPK} = \beta_k \times \frac{\text{sales revenue}}{\text{value of plant/mach}}$. Panel B presents the borrowing and liquidity effects in the high-MRPK triple difference estimation when the industry is limited to manufacturing only, where capital is more reliably measured. Error bars present 95% confidence interval and standard errors are clustered by district.

Figure A.11: Case Type Distribution
Panel A: Case Type Distribution in Matched Firm Data



Panel B: IBC-Adjacent vs Non-Adjacent Cases in High Judge Capacity Over Time



Notes: Panel A presents percentage of cases by type in the matched-firm case-level data from [Rao 2025](#). [Ash et al. 2025](#) does not provide the identity of litigants to generate matched-firm case-level data. Panel B uses case-level data from [Ash et al. 2025](#) to calculate the number of IBC-adjacent such as money suits, recovery suits, debt arbitration, challenges to possession, execution petitions, and placebo (criminal) cases over time. The graph presents event-study estimates of IBC reform by district judicial capacity in a DiD specification with district, year, and state-year fixed effects. Standard errors are clustered by district.

Table A.1: Firm Type Across High and Low Judge-Pop Ratio Districts

Panel A						
	High Default Risk			Low Default Risk		
	High Judge-Pop	Low Judge-Pop	P-val	High Judge-Pop	Low Judge-Pop	P-val
Overall Credit Rating	2.321 (0.916)	2.342 (0.897)	0.763	6.313 (1.399)	6.009 (1.354)	0.006
Debt to Asset	0.459 (0.351)	0.570 (0.441)	0.929	0.375 (0.342)	0.401 (0.330)	0.319
Total Assets (Deflated)	3341.437 (5561.150)	2468.327 (3934.879)	0.148	30554.326 (164600.758)	13844.836 (45273.843)	0.000
<i>N</i>	159	374		1643	2971	

Panel B						
	High MRPK			Low MRPK		
	High Judge-Pop	Low Judge-Pop	P-val	High Judge-Pop	Low Judge-Pop	P-val
Overall Credit Rating	6.091 (1.719)	5.862 (1.641)	0.758	5.972 (1.857)	5.575 (1.812)	0.095
Debt to Asset	0.319 (0.269)	0.349 (0.281)	0.006	0.514 (0.484)	0.526 (0.458)	0.850
Total Assets (Deflated)	20241.117 (103905.870)	10511.783 (33645.469)	0.000	46931.853 (227576.917)	17167.062 (54591.554)	0.001
<i>N</i>	740	1162		833	1702	

Notes: This table presents debt-related characteristics of high and low default risk firms (Panel A) and high and low MRPK firms (Panel B) in high and low judge-population ratio districts using data from pre-2010. Overall credit rating ranges from 1 to 8, with 1 indicating lowest possible rating and 8 indicating highest. Debt to Asset is total debt (borrowing) to total asset ratio and total asset is measured in Million INR in 2015 levels.

Table A.2: Unobserved Firm Exits: Missing Financial Variables in Prowess Data Over Time

	(1) Firm Missing Prowess All	(2) Firm Exit Prowess All	(3) Firm Missing Prowess Default-Risk	(4) Firm Exit Prowess Default-Risk	(5) Firm Missing Prowess MRPK	(6) Firm Exit Prowess MRPK
High Jdg Pop Ratio x Post	-0.0113 (0.00903)	-0.00743 (0.00633)	0 (.)	0 (.)	0 (.)	0 (.)
High Jdg Pop Ratio x High Default Risk x Post			-0.0434* (0.0236)	-0.0286 (0.0281)		
High Default Risk x Post			0.0187 (0.0196)	0.0429*** (0.0146)		
High Jdg Pop Ratio x High MRPK x Post					-0.00764 (0.0159)	-0.00251 (0.0132)
High MRPK x Post					-0.0123 (0.0125)	0.00378 (0.0130)
Observations	64275	64275	63060	63060	63060	63060
No. Districts	242	242	165	165	165	165
Firm FE	X	X	X	X	X	X
District-Year FE	No	No	X	X	X	X
Industry-Year FE	X	X	X	X	X	X
Mean Dep Var	0.310	0.0300	0.310	0.0300	0.310	0.0300
SD Dep Var	0.460	0.170	0.460	0.170	0.460	0.170
Adj R-Squared	0.772	0.699	0.774	0.701	0.774	0.701

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table is the DiD/DDD equivalent to [Figure A.4](#) using firm disappearance in the financial data as the dependent variable. I use all firms observed in Prowess data on or before 2010 and generate indicators for missing financial data to assess whether the interaction between bankruptcy reform and local district judicial capacity affects reporting compliance by firms. Column 1 reports DiD estimates examining the interaction between high judicial capacity and the timing of bankruptcy reform using all firms. Columns 2 and 3 report DDD estimates using the interaction between firm characteristic, district judicial capacity, and the timing of the reform. Standard errors are clustered by district.

Table A.3: Selection into Reporting: Financial Variables in Prowess Data Over Time

	(1) Firm Reports Prowess All	(2) Firm Reports Prowess Default-Risk	(3) Firm Reports Prowess MRPK
High Jdg Pop Ratio x Post	0.0113 (0.00903)	0 (.)	0 (.)
High Jdg Pop Ratio x High Default Risk x Post		0.0434* (0.0236)	
High Default Risk x Post		-0.0187 (0.0196)	
High Jdg Pop Ratio x High MRPK x Post			0.00764 (0.0159)
High MRPK x Post			0.0123 (0.0125)
Observations	64275	63060	63060
No. Districts	242	165	165
Firm FE	X	X	X
District-Year FE	No	X	X
Industry-Year FE	X	X	X
Mean Dep Var	0.690	0.690	0.690
SD Dep Var	0.460	0.460	0.460
Adj R-Squared	0.772	0.774	0.774

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table is the DiD/DDD equivalent to [Figure A.4](#) using firm disappearance in the financial data as the dependent variable. I use all firms observed in Prowess data on or before 2010 and generate indicators for missing financial data to assess whether the interaction between bankruptcy reform and local district judicial capacity affects reporting compliance by firms. Column 1 reports DiD estimates examining the interaction between high judicial capacity and the timing of bankruptcy reform using all firms. Columns 2 and 3 report DDD estimates using the interaction between firm characteristic, district judicial capacity, and the timing of the reform. Standard errors are clustered by district.

Table A.4: Bank-Firm Relationship

	DiD (1) Bank-Firm Relation	(2) No. Banks	Triple-D (3) Bank-Firm Relation	(4) No. Banks	Triple-D (5) Bank-Firm Relation	(6) No. Banks
High Jdg Pop Ratio x Post	0.00249 (0.0125)	-0.327 (0.265)	0 (.)	0 (.)	0 (.)	0 (.)
High Jdg Pop Ratio x High Default-Risk x Post			0.0160 (0.0297)	0.188 (0.622)		
High Default-Risk x Post			-0.0171 (0.0249)	0.814* (0.433)		
High Jdg Pop Ratio x High MRPK x Post					0.0291 (0.0264)	1.482 (1.120)
High MRPK x Post					0.00530 (0.0185)	-0.0915 (0.602)
Observations	343620	343620	343290	343290	343290	343290
No. Districts	236	236	214	214	214	214
Firm FE	X	X	X	X	X	X
District-Year FE	N	N	X	X	X	X
Bank FE	X	X	X	N	X	N
Mean Dep Var	0.380	6.760	0.380	6.760	0.380	6.760
SD Dep Var	0.490	8.370	0.490	8.370	0.490	8.370
Adj R-Squared	0.240	0.666	0.253	0.695	0.253	0.695

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table is equivalent to [Figure A.6](#), depicting the effect of the reform on firm-bank relationship observed in the “banker” table corresponding to the Prowess firm sample. There are two relevant margins of this relationship: first, whether there exists a relationship between a firm and a bank and second, total number of banking relationships by a firm. There are 658 unique banks with at least one relationship with the sample firm. The banks span commercial banks as well as regional banks and credit unions. The dependent variable “Existence of Bank-Firm Relationship” is a dummy variable that takes value 1 when I observe a relationship between a specific bank and a sample firm for a given year and 0 otherwise. The variable “Number of Bank-Firm Relationships” is constructed by aggregating the existence indicator across all banks for a specific firm for a given calendar year. The data includes a balanced panel of all sample firms over the study period 2010-2024. Additionally, the existence indicator also includes the specific banking partner for the firm. Col 1-2 present DiD estimates comparing the outcomes between high judicial capacity districts with those in low capacity districts. For regressions using the existence variable, I also include bank fixed effects in addition to firm and district-year fixed effects. Regressions using the aggregate variable includes only firm and district-year fixed effects. Col 3-6 present the triple difference event study results by firms classified as high default risk or high MRPK, respectively, and the underlying judicial capacities of the district. Standard errors are clustered by district.

Table A.5: Bankruptcy Reform and District Judicial Capacity: All Firms

	(1) Arc Sine Borrowing	(2) Arc Sine Net Liquidity	(3) Arc Sine Legal Expenditure	(4) Arc Sine Interest Expenditure	(5) Arc Sine Investment	(6) Arc Sine Sales Revenue
High Jdg Pop Ratio x Post	0.0967*** (0.0348)	-0.134 (0.106)	0.0379** (0.0185)	0.0953*** (0.0236)	-0.0231 (0.0227)	-0.0122 (0.0218)
Observations	59025	59070	51405	58395	45825	58560
No. Districts	240	240	228	239	215	239
Firm FE	X	X	X	X	X	X
Industry-Year FE	X	X	X	X	X	X
Mean Dep Var	7.400	2.360	2.870	5.050	5.330	8.650
SD Dep Var	2.200	6.520	1.720	1.830	2.920	1.870
Adj R-Squared	0.803	0.626	0.906	0.803	0.878	0.872

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from DiD specification using time periods before and after the 2016 reform across high and low-judicial-capacity districts on borrowing and production outcomes of all firms in the sample districts. High-judge-to-population ratio districts are those with the ratio above the sample average. Standard errors are clustered by district.

Table A.6: Using Alternate Cut-off For High Judicial Capacity: Firm Borrowing and Liquidity

	(1) Arc Sine Borrowing	(2) Arc Sine Net Liquidity	(3) Arc Sine Borrowing	(4) Arc Sine Net Liquidity
Above Median Jdg-Pop x High Default Risk X Post	-0.213** (0.0934)	0.471 (0.429)		
High Default Risk X Post	0.135* (0.0710)	-0.567** (0.242)		
Above Median Jdg-Pop x High MRPK X Post			0.0115 (0.0579)	0.485* (0.270)
High MRPK x Post			0.0521 (0.0551)	-0.655** (0.264)
Observations	57840	57885	47970	48000
No. Districts	164	164	150	150
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	7.400	2.360	7.480	2.060
SD Dep Var	2.200	6.520	2.290	6.700
Adj R-Squared	0.801	0.624	0.797	0.616

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from [Equation 1](#) using a different definition of high judicial capacity. Specifically, I use median as the cut-off instead of average. Standard errors are clustered by district.

Table A.7: Using Alternate Cut-off For High Judicial Capacity: Production Outcomes of High Risk Firms

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
Above Median Jdg-Pop x High Default Risk X Post	-0.0460 (0.0555)	-0.147* (0.0785)	-0.254 (0.155)	-0.0290 (0.0916)
High Default Risk X Post	-0.0392 (0.0493)	0.0352 (0.0553)	0.0294 (0.103)	-0.0515 (0.0650)
Observations	50190	57210	44700	57360
No. Districts	152	164	145	163
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.870	5.050	5.330	8.650
SD Dep Var	1.720	1.830	2.920	1.870
Adj R-Squared	0.906	0.805	0.878	0.871

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using a different definition of high judicial capacity. Specifically, I use median as the cut-off instead of average. Standard errors are clustered by district.

Table A.8: Using Alternate Cut-off For High Judicial Capacity: Production Outcomes of High MRPK Firms

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
Above Median Jdg-Pop x High MRPK X Post	-0.0148 (0.0594)	0.0292 (0.0631)	0.244** (0.119)	0.0585 (0.0506)
High MRPK x Post	-0.0156 (0.0303)	-0.0259 (0.0506)	-0.0588 (0.0972)	-0.159*** (0.0437)
Observations	41070	47400	36795	47520
No. Districts	136	150	130	149
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.910	5.140	5.540	8.690
SD Dep Var	1.760	1.890	2.950	1.940
Adj R-Squared	0.905	0.801	0.877	0.873

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using a different definition of high judicial capacity. Specifically, I use median as the cut-off instead of average. Standard errors are clustered by district.

Table A.9: Using Pre-Reform Judge Vacancy: Firm Borrowing and Liquidity

	(1) Arc Sine Borrowing	(2) Arc Sine Net Liquidity	(3) Arc Sine Borrowing	(4) Arc Sine Net Liquidity
Below Median Vacancy x High Default Risk X Post	-0.107 (0.117)	0.398 (0.342)		
High Default Risk X Post	0.0938 (0.0759)	-0.566** (0.232)		
Below Median Vacancy x High MRPK X Post			0.0202 (0.0648)	0.255 (0.269)
High MRPK x Post			0.0506 (0.0566)	-0.536*** (0.205)
Observations	57600	57645	47730	47760
No. Districts	162	162	148	148
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	7.400	2.360	7.480	2.060
SD Dep Var	2.200	6.520	2.290	6.700
Adj R-Squared	0.801	0.624	0.796	0.616

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using a different definition of high judicial capacity. Specifically, I use below median pre-reform judge vacancy rate as the cut-off instead of judge-to-population ratio. Standard errors are clustered by district.

Table A.10: Using Pre-Reform Judge Vacancy: Production Outcomes of High Risk Firms

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
High Default Risk x Post	-0.0525 (0.0480)	-0.0299 (0.0601)	-0.0630 (0.154)	-0.0774 (0.0607)
Below Median Vacancy x High Default Risk x Post	-0.0185 (0.0585)	-0.00775 (0.0968)	-0.0477 (0.148)	0.0242 (0.0891)
Observations	49995	56970	44490	57120
No. Districts	150	162	143	161
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.870	5.050	5.330	8.650
SD Dep Var	1.720	1.830	2.920	1.870
Adj R-Squared	0.906	0.805	0.878	0.871

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using a different definition of high judicial capacity. Specifically, I use below median pre-reform judge vacancy rate as the cut-off instead of judge-to-population ratio. Standard errors are clustered by district.

Table A.11: Using Pre-Reform Judge Vacancy: Production Outcomes of High MRPK Firms

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
High MRPK x Post	-0.0104 (0.0429)	-0.0323 (0.0390)	0.0783 (0.0973)	-0.175*** (0.0423)
Below Median Vacancy x High MRPK X Post	-0.0229 (0.0642)	0.0490 (0.0605)	-0.0277 (0.143)	0.0875** (0.0437)
Observations	40875	47160	36585	47280
No. Districts	134	148	128	147
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.910	5.140	5.540	8.690
SD Dep Var	1.760	1.890	2.950	1.940
Adj R-Squared	0.906	0.801	0.877	0.873

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from [Equation 1](#) using a different definition of high judicial capacity. Specifically, I use below median pre-reform judge vacancy rate as the cut-off instead of judge-to-population ratio. Standard errors are clustered by district.